Cardiovascular Disease Detection Using Deep Learning: A CNN-Based Ensemble Approach

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FINAL YEAR DESIGN PROJECT REPORT

This Report Presented in Partial Fulfillment of the Requirements for the **Degree of Bachelor of Science in Computer Science and Engineering**

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January 06, 2026

APPROVAL

This Project titled **Cardiovascular Disease Detection Using Deep Learning: A CNN-Based Ensemble Approach**, submitted by S. M. Tanvir Ahmed, ID No: **221-15-5011** and Avinandan Roy, ID No: **221-15-4899** to the Department of Computer Science and Engineering, Daffodil International University has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents. The presentation has been held on **6 January, 2026**.

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ACKNOWLEDGEMENTS

This work would not have been possible without the support and contributions of many individuals over the past two semesters. We are deeply grateful to everyone who has as- sisted us in one way or another.

First, we express our heartfelt thanks and gratefulness to the almighty for His divine blessing making it possible for us to complete the **Final Year Design Project(FYDP)** successfully.

We are grateful and wish our profound indebtedness to **Md. Jakaria Zobair**, **Lecturer**, Department of Computer Science and Engineering, Daffodil International University, Dhaka, Bangladesh. Deep knowledge and keen interest of our supervisor in the field of **Deep Learning to** carry out this project. His endless pa- tience, scholarly guidance, continual encouragement, constant and energetic supervision, constructive criticism, valuable advice, reading many inferior drafts, and correcting them at all stages have made it possible to complete this project.

We would like to express our heartfelt gratitude to the Head of the Department of Com- puter Science and Engineering, for his kind help in finishing our project and also to other faculty members and the staff of the Department of Computer Science and Engineering, Daffodil International University.

We would like to thank our entire course-mates at Daffodil International University, who took part in this discussion while completing the coursework.

Finally, we must acknowledge with due respect the constant support and patience of our parents.

ABSTRACT

CVDs are a huge health burden in the world, thus the necessity to have precise and automated diagnostic systems. The paper will suggest a deep learning-based cardiovascular disease detection system based on cardiac magnetic resonance imaging (MRI) and focus on the enhancement of classification performance through ensemble learning. There are four models of transfer learning-based convolutional neural network ResNet50, VGG16, DenseNet121, and EfficientNetB0, which are used to classify images at the levels of images after standardized preprocessing and data augmentation. To achieve robustness and generalization, the predictive results of individual models are combined with average, weighted and voting ensemble methods. Model interpretability Gradient-weighted Class Activation Mapping (Grad-CAM) handles model interpretability, which offers a visual explanation of the area of decision-relevance in cardiac MRI data. The experimental test shows that ensemble learning is always better than the single CNN models and the weighted ensemble is the best with the highest classification accuracy of 97.36. The optimized ensemble model is further geared towards deployment of mobile applications, which suggests that it can be used practically in the clinical assistance. In general, the given solution can be discussed as suitable and valid, interpretable, and deployable to detect cardiovascular disease using cardiac MRI data through the proposed approach.

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Chapter 1

Introduction

This chapter presents the general focus of the research and the reasons as to why automated cardiovascular disease detection is a critical issue to consider. It introduces the rationale of the application of deep learning and medical imaging to the early diagnosis, and the primary goals that lead the study. The chapter describes the methodology adopted in the creation of the CNN-based ensemble system and gives a preview of the most important results obtained during the project. It also talks about the way the report is structured to assist the reader to know the flow of the work. The chapter establishes a foundation on which the finer discussions are to be said since it allows us to establish the context and the purpose of the thesis. All in all, it defines the significance, scale, and the purpose that the proposed system is supposed to serve.

1.1 Introduction

Cardiovascular diseases (CVDs) are considered one of the most frequent causes of mortality among the global population, annually impacting millions of people and causing excessive pressure on the healthcare sector. Early diagnosis and proper diagnosis is very important because early diagnosis is very important as far as treatment outcome and long term survival are concerned. In the contemporary clinical practices, there are multiple imaging techniques, including ECG, echocardiography, chest X-ray agglomeration, and cardiac MRI, that are utilized to diagnose heart structure and abnormal functioning. The significance of medical images in the identification of conditions, such as arrhythmia, myocardial infarction, cardiomyopathy, and other structural heart conditions has been emphasized by recent research and justifies the increasing popularity of reliable and automated diagnostic tools [1], [2], [4].

The development of deep learning has demonstrated the impressive capabilities in the field of medical image analysis, especially the transfer learning and convolutional neural networks (CNNs). VGG16, ResNet50, DenseNet, EfficientNet and MobileNet are some of the models that have been applied successfully on ECG signals, MRI images, echocardiograms, and X-rays to provide high diagnostic accuracy in various cardiac conditions [2], [3], [6], [7], [17]. It has also been shown that hybrid methods combining segmentation networks such as U-Net or DeepLabv3, optimization based feature

selection and ensemble learning outperform poorly without this integration [7], [13], [24], [25]. Specifically, transfer learning has also allowed researchers to learn potent pre-trained CNNs on medical imaging tasks with small data set, rendering it a highly-used method in CVD detection [10], [11], [29].

In general, relating this technological advancement to clinical requirements, deep learning is a perspective solution to automation of cardiovascular disease detection and more accurate diagnostics. Nevertheless, biased CNN models can tend to score differently on various datasets, which inspires the application of ensemble methods in order to synthesize the advantages of each. This thesis is developed based on that concept whereby an ensemble model is developed using CNN to identify cardiovascular disease based on MRI images. The system employs four transfer-learning models, i.e., VGG16, ResNet50, EfficientNet, and DenseNet, to generate different feature representations and fuse them with the help of averaging, weighted fusion, and majority voting to improve the degree of classification reliability. The methodological pipeline, consisting of preprocessing images, augmentation, feature analysis, and comprehensive evaluation, is expected to develop powerful and generalizable tool that will be compatible with the continuing trend of using deep learning to aid in the early and precise heart disease detection [3], [7], [24].

1.2 Motivation

The cardiovascular diseases still are the significant challenge to the whole world, and still, there is a necessity to detect them in the early stages to avoid serious consequences and mortality. Clinicians are using a number of imaging modalities, including ECG, X-ray, echocardiography, and MRI; however, the interpretation might be time-consuming and depend on the clinical expertise. Recent publications have also reported great rates of improvement in diagnostic performance through deep learning networks such as ResNet50, VGG16, DenseNet, and EfficientNet to detect arrhythmia, myocardial infarction, cardiomyopathy, and structural abnormalities [1], [2], [4], [7]. Nevertheless, the limitation of the diversity of datasets, a lack of real-world validation, and the aspect of intermittent performance of single CNN models under different imaging aspects are also pointed in these studies [1], [3], [11]. These deficiencies drive the need to have more powerful approaches, which are better able to generalize and offer reliable assistance in clinical judgment.

Meanwhile, ensemble learning has been successful in enhancing the accuracy of the classification using the strengths of various deep learning models. Previous studies

prove that hybrid or ensemble methods usually perform better than single architectures and provide more robust predictions on diverse data [24], [25]. This poses a significant desire to investigate ensemble based approaches to cardiovascular disease detection particularly through MRI images, which provide precise structural detail of the heart. This thesis will create a more acceptable detection system by combining four strong transfer-learning models, including VGG16, ResNet50, EfficientNet, and DenseNet, to gain the advantages of different representations on features and better generalization. The rationale of the work is to design a viable, precise, and replicable deep learning architecture that can be consistent with the current attempts to improve automated CVD diagnosis and reinforce clinical activities.

1.3 Objectives

The cardiovascular disease detection system created based on the deep learning method applied to cardiac MRI images and their accurate and reliable analysis is the main goal of the given research. The study will specifically focus on systematic testing of several transfer learning models and subsequent improvement of them with the help of ensemble learning, whereas the consideration of their practical usability by visual explainability and mobile application will be a point of concern. The objectives of this thesis are specific and they are as follows:

1. The aim of the project to create a deep learning-based cardiovascular disease detector using cardiac MRI images - This task aims at creating a structured pipeline that takes cardiac MRI images and then performs image-level classification between normal and diseased conditions.
2. To compare the performance of various transfer learning CNN models - The research will employ and compare ResNet50, VGG16, DenseNet121, and EfficientNetB0 when all models are trained and tested in a single environment to determine their respective advantages and disadvantages.
3. To enhance the accuracy of the diagnosis with the help of ensemble learning methods - The aim of this task is to combine several CNN models with the help of the average, voting, and weighted ensemble methods to increase the robustness of the classification and decrease the number of mistakes during the prediction.
4. To achieve the interpretability of models through visual explanation methods - The study aims to use Grad-CAM visualizations to distinguish valuable areas in cardiac MRI images to facilitate transparent and clinically meaningful decision-making.
5. To prove usefulness in the form of deployment of mobile applications - The last task

is to implement the optimized ensemble model in a mobile application setting allowing the accessible and real-time detection of cardiovascular diseases to be used practically.

1.4 Methodology

The workflow of this thesis is a sequential, step-by-step methodology that aims at creating a robust deep learning-based cardiovascular disease detector of MRI images. It starts with the stage of environmental preparation, library configuration, and dataset verification to make sure that all the necessary elements to train and evaluate are configured. It then systematizes the data, which includes the Normal and Sick MRI images and divides it into training, validation, and testing data sets then proceeds to augment the images to decrease overfitting and enhance the overall performance. Following the preprocessing step, four transfer-learning CNNs, namely, VGG16, ResNet50, EfficientNet, and DenseNet are built upon by freezing the trained weights and adding new custom layers that are used to perform binary disease classification.

All the models are individually trained on augmented images, and to stabilize and optimize the training, early stopping, scheduling of the learning-rate and checkpointing are used. After the individual performance measures have been gathered, three ensemble methods, which include simple averaging, weighted averaging using AUC and majority voting are applied to integrate the predictive abilities of the four CNN models. These ensembles are assessed on the test set in terms of accuracy, precision, recall, F1-score, ROC-AUC and confusion matrices that enable a detailed comparison with the respective models. The last part of the methodology is to visualize the Grad-CAM on the randomly selected test images to determine the behavior of the classification in the real world. It is by using this systematic approach that the methodology can guarantee that the end ensemble model is robust and able to provide better cardiovascular disease detection performance.

1.5 Project Outcome

The results of this project attest to the possibilities of deep learning and its ability to assist in the early and dependable detection of cardiovascular diseases with the help of MRI images. With structural differences between Normal and Sick cardiac scans, the system will provide a data-based methodology that could assist in reducing the diagnostic uncertainty and aid the clinical decision-making. The findings underscore the importance of automated analysis of images in supplementing conventional medical

evaluation particularly in environments where the expertise is scarce or the volume of work is very high.

In the methodological perspective, four transfer-learning CNN models, i.e., VGG16, ResNet50, EfficientNet, and DenseNet, are effectively combined into a single ensemble model. The models are learnt by different patterns on the MRI images, and the concerted efforts of both models produce better accuracy, precision, recall, and AUC than when each model is used separately. The Grad-CAM based visualizations of the feature also affirm that the models acquire meaningful and well separated representations of the Normal and Sick cases. The most stable and reliable predictions were obtained using the ensemble strategies, especially the weighted averaging, which justifies the importance of combining various deep learning models.

The developed system may be useful in practice to help clinicians by providing a fast and automatic second opinion of MRI scans. The results of the prediction, as well as the scores of confidence and the visual interpretation, should be used to prioritize those cases which might need additional investigation. These tools can be particularly beneficial in hospitals where there is a low amount of radiology staff or in telemedicine systems where the specialists are not necessarily on hand. Also, the system can be expanded by adapting the workflow to other datasets of cardiac imaging.

1.6 Organization of the Report

The thesis is structured into six chapters in which each of the chapters has a major part of the research in a clear and systematic way. The layout facilitates the reader in tracking the flow of the work- starting with an insight into the problem space to the discussion of the methodology, implementation, and concluding findings without fail in line with the overall objectives of the study of cardiovascular disease detection with the help of deep learning.

Chapter 1: Introduction - This chapter presents the research problem, the reasons that led to the creation of the deep learning-based cardiovascular disease detection system, and indicates the aims of the research. It also provides the overview of the proposed methodology, defines the key findings, and provides an overview of the structure of the report.

Chapter 2: Background - Chapter 2 provides the required theoretical framework and the literature review regarding the previous research terms in regard to cardiovascular disease detection with the use of CNNs, transfer learning, ensemble models, and medical

imaging methods. It contains a comprehensive literature review regarding the works mentioned and pinpoints the existing research gaps that will be taken care of with the help of this thesis.

Chapter 3: Research Methodology - The approach to methodology is described in this chapter. It explains the data employed, data preprocessing, and data augmentation techniques, and the four models of transfer learning architecture. The ensemble strategies, evaluation metrics, the steps of Grad-CAM visualization and the general system workflow that defines the development process are also outlined in the chapter.

Chapter 4: Implementation and Results - Chapter 4 gives an account of how the proposed system is implemented, such as setting up the environment, training the model, and hyperparameter settings. It gives a thorough analysis of single CNN models and ensemble techniques in terms of accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices. To support the results, visual illustrations of predictions and separability of features are provided.

Chapter 5: Engineering Standards and Design Challenges - In this chapter, the author talks about engineering standards to be used in the research process, such as software practices, computational constraints, and ethical considerations. It also looks at the design issues that have been faced during the development process including how to deal with overfitting, data constraints, and model selection and the remedies that have been put in place to deal with such issues.

Chapter 6: Conclusion - The last chapter is a conclusion of the main findings and results of the research. It also puts emphasis on the advantages of the ensemble-based approach to detecting cardiovascular disease and the shortcomings of the research. It also offers suggestions on how to work in the future, such as expanding the diversity of datasets, increasing real-world clinical validation, and the expansion of the deployment features of the system.

Chapter 2

Background

This chapter introduces the background knowledge that is required to comprehend the methods and principles that were adopted in the development of the cardiovascular disease detection system. It discusses the key areas that include deep learning, convolutional neural networks, medical imaging, and transfer learning. A comprehensive literature review has been presented in order to emphasize on past research, research findings and limitations that have led to this study. The most significant gaps in the current practices are also found in the chapter as they demonstrate the reason as to why an ensemble-based approach to deep learning is useful. The chapter provides a very good theoretical foundation of the rest of the thesis by connecting prior work to the suggested methodology. In general, it makes the reader ready to comprehend how the project expands, enhances, and diverges ahead of other works.

2.1 Introduction

Cardiovascular diseases are among the most severe health problems in the world, as they have become the cause of great mortality and long-term effects. Early diagnosis is a crucial component of preventing the course of the disease, but when the interpretation is made through conventional means and in particular, with medical imaging, the process requires the expertise of an expert and can be quite time-intensive. The recent developments in the field of artificial intelligence have seen the rise of deep learning as an effective method of medical image analysis and diagnosing subtle patterns that might be difficult to distinguish by a clinician. CNNs specifically have demonstrated good results in the detection of anomalies in ECG signals, X-ray images, echocardiograms and MRI scans, and thus are viable options when it comes to automated cardiovascular disease detection.

This chapter imparts the background knowledge one needs to know about the application of deep learning and transfer learning on this thesis. It presents the main concepts, image preprocessing, CNN architectures, feature extraction, and ensemble learning, which are actively discussed in the recent research to enhance the accuracy of disease classification. The literature review is also thoroughly provided in the chapter and

summarizes the works of other researchers, who have used a model such as ResNet50, VGG16, DenseNet, EfficientNet, and other ensemble systems to diagnose the heart. The comparison of these works sheds light in the chapter on the shortcomings of existing methods, especially in terms of model generalization, model diversity, and interpretability.

In the end, this background section sets the basis of reasons why an ensemble-based deep learning framework will be developed to detect cardiovascular disease using MRI. It examines the gaps in the research by reviewing the relevant theories and past applications, which helps it to establish the research gaps that this study seeks to fill in and gives the scientific context on how to interpret the methodology and contributions of the proposed system.

2.2 Literature Review

Gulhane & Kumar [1] propose a deep learning-based model using ResNet50 to detect heart diseases from ECG images. Trained on publicly available Kaggle datasets, the model achieved a validation accuracy of 91.88%, improving to 94.27% with further training. The model was designed to detect conditions such as arrhythmia, myocardial infarction (MI), and previous MI (PMI). However, the study does not explore the model's performance under real-world clinical conditions or test its generalizability across different datasets.

Maqbool et al. [2] utilize CNNs and ResNet50 for detecting cardiac diseases from ECG images. The model demonstrated a high accuracy of 99.12% in categorizing normal and abnormal heart conditions (MI, AHB, PMI). This approach performs well for ECG image classification, but the study does not address the robustness of the model when applied to different patient demographics or across varied datasets.

Mahmoud et al. [3] propose a modified ResNet50 model for cardiovascular disease (CVD) prediction using ECG signals. Their results showed a 4.83% improvement in accuracy compared to the original ResNet50 model. The modified version outperformed LeNet-5 in detecting CVDs. However, the study does not explore the potential for other feature extraction methods or alternative CNN architectures that might enhance model performance.

Xu et al. [4] present a fine-tuned ResNet50 model to identify hypertrophic cardiomyopathy (HCM) and dilated cardiomyopathy (DCM) from ECG images. The model demonstrated strong performance with an AUROC of 0.996 for DCM and 0.980 for HCM detection. The fine-tuned ResNet50 model is promising for early detection;

however, the study does not investigate its effectiveness across larger and more diverse datasets or evaluate its performance in real-time clinical use.

Yu & Chen [5] integrate machine learning with medical imaging to improve cardiac disease detection. Their approach combines echocardiography, MRI, and chest radiographs with machine learning classifiers, achieving a diagnostic accuracy of 96%. This hybrid method outperforms models based only on clinical data. However, the study lacks an evaluation of the computational cost and real-time feasibility of integrating multiple imaging modalities.

Anand et al. [6] enhance arrhythmia detection using an improved ResNet-50 deep learning model trained on the MIT-BIH arrhythmia dataset. The model achieved 98.14% accuracy, demonstrating the potential of ResNet-50 for automatic arrhythmia detection. However, the study does not discuss the model's applicability in real-world clinical environments or its scalability to larger, more diverse datasets.

Mohanapriya et al. [7] propose a deep learning-based method using DeepLabv3 for semantic segmentation and ResNet for feature extraction to detect myocardial abnormalities in MRI scans. The model outperforms traditional segmentation methods, showing improved accuracy in identifying important cardiac structures. However, the study does not assess the model's generalizability across various myocardial diseases or patient demographics.

Hemalatha et al. [8] compare Capsule Networks (CapsNet) and VGG16 for early cardiac disease prediction. VGG16 outperforms CapsNet with an accuracy of 99.8%, demonstrating its potential for improved heart disease detection. However, the study does not explore the possible benefits of combining CapsNet with other deep learning techniques for enhanced prediction accuracy.

Raju et al. [9] implement federated learning with VGG16 for ECG-based heart attack detection. This privacy-preserving model achieved promising results and improved heart attack prediction accuracy. However, the study does not evaluate the model's performance on diverse datasets or assess the computational complexity of federated learning in ECG machines.

Addisu et al. [10] introduce a hybrid machine-learning approach that combines transfer learning using VGG16 with various machine-learning classifiers for heart disease detection. The VGG16-Random Forest hybrid achieved 92% accuracy, providing valuable support in clinical decision-making. Additionally, the study integrates explainable AI techniques, such as SHAP values, to interpret the model's predictions. However, the study does not evaluate the model's robustness across different healthcare settings or its

computational efficiency.

Van-Binh & Thai-Nghe [11] propose a transfer learning approach to detect cardiovascular diseases, specifically cardiomegaly and aortic aneurysm, using X-ray chest images. The model, based on the pre-trained VGG16 deep learning model, achieved accuracies of 0.95, 0.96, and 0.70 in three experimental scenarios. This method demonstrates a strong approach to detecting cardiovascular conditions from X-ray images. However, the study does not evaluate the model's performance on more diverse datasets or assess its robustness under real-world conditions.

Boulares et al. [12] apply transfer learning to a public dataset for cardiovascular disease (CVD) recognition from phonocardiogram (PCG) signals. The study provides an experimental benchmark for future research in CVD detection from PCG signals. While the study offers a valuable benchmark for the field, it lacks denoising or cleaning steps, and the results may be influenced by the dataset's inherent noise.

Krishnan et al. [13] propose a hybrid optimization-based feature selection method using the DenseNet model for heart disease prediction. By combining Lion Algorithm (LA) and Butterfly Optimization Algorithm (BOA), the model achieved improved classification accuracy. The proposed approach demonstrates improved accuracy and reduced training time by selecting the most relevant features for CVD prediction. However, the study does not compare the hybrid method with other feature selection techniques or evaluate the model's robustness in various clinical settings.

Mandava [14] presents the MDensNet201-IDRSRNet, a hybrid deep learning system for efficient cardiovascular disease prediction. The model utilizes Modified DenseNet201 for feature extraction and an Improved Deep Residual Shrinkage Network (IDRSNet) for CVD prediction. The model achieved an accuracy of 99.12% on benchmark datasets. However, the study lacks a detailed analysis of the system's computational complexity and its ability to generalize across diverse healthcare settings.

Sivanathbabu & Kamalakkannan [15] propose an optimized DenseNet model for heart disease prediction using medical health records. The model, trained on a Kaggle dataset of 70,000 cases, achieved an accuracy of 91.86%. The proposed model outperforms other existing models in prediction accuracy, making it a promising tool for heart disease detection. However, the study does not explore the impact of different hyperparameters on model performance or evaluate its effectiveness on diverse datasets.

Aborokbah [16] applies EfficientNet to Alzheimer's disease MRI classification. The study used region-growing methods and U-Net for segmentation, followed by EfficientNet-B0 for classification. The model achieved a high accuracy of 98.12%, demonstrating the

effectiveness of EfficientNet for Alzheimer's detection. However, the study does not explore the model's ability to detect other forms of dementia or its performance on datasets with varying levels of quality.

Mahtab et al. [17] enhance heart disease detection in echocardiogram images using an optimized EfficientNetB3 model. The model achieved a test accuracy of 94.53% after fine-tuning and experimentation. The proposed model shows great potential for early heart disease detection. However, the study does not address the computational cost of the model or its performance in a real-world clinical environment.

Mirsharifi & Keyvanpour [18] introduce an EfficientNet-based method for interpretable early detection of Alzheimer's disease. The study used the ADNI dataset and achieved a classification accuracy of 97.84%, improving upon previous methods. GradCAM++ was applied for interpretability, highlighting brain regions influencing the model's predictions. However, the study does not evaluate the model's effectiveness across different stages of Alzheimer's or test it in diverse patient populations.

Dhanalaxmi et al. [19] propose a novel MobileNetV3-based method for cardiovascular disease classification. The model achieved an accuracy of 99.82% for the UCI Heart Disease dataset and 99.73% for the Framingham dataset. The MobileNetV3 approach outperforms traditional learning techniques. However, the study does not explore the model's scalability or its performance on other cardiovascular datasets.

Priyadharshini & Madheswaran [20] introduce CVD-TransNet, a transfer learning-based method for cardiovascular disease classification using ECG signals. The model achieved an impressive accuracy of 99.62%, surpassing other methods like Deep CNN and SVM. The study highlights the effectiveness of the proposed model for early CVD detection. However, the study does not assess the model's robustness under noisy or incomplete data conditions.

Willis, Zhou, & Liu [21] propose using deep learning methods for cardiovascular disease detection and risk prediction from fundus images. The study uses convolutional neural networks (CNNs) and transfer learning to detect cardiovascular diseases from retinal images, showing promising results with the potential to assist in routine eye exams. However, the study does not explore the generalization of the model to different patient populations or other imaging modalities.

Sayin et al. [22] develop a CNN model based on the InceptionV3 architecture for detecting myocardial infarction (MI) from ECG images. The model, trained using data from the Ch. Pervaiz Elahi Institute of Cardiology, achieved 93.27% accuracy. The study demonstrates the feasibility of deep learning models for detecting CVD from ECG

images, but it does not assess the model's performance across different clinical datasets or patient demographics.

Gajendran et al. [23] apply transfer learning to classify ECG patterns associated with cardiac arrhythmias. The research utilizes deep neural networks pretrained on ImageNet for classifying ECG scalograms, showing promising results in ECG time-series classification. However, the study does not compare the transfer learning models with traditional machine learning methods for ECG classification in a comprehensive manner.

Baccouche et al. [24] propose an ensemble deep learning framework for heart disease classification using a dataset from Mexico. The model, combining BiLSTM, BiGRU, and CNN, achieved an accuracy between 91% and 96% for different heart disease types. The study addresses the issue of class imbalance in heart disease datasets and demonstrates the potential of ensemble learning to improve classification accuracy. However, it does not explore the model's performance on larger, more diverse datasets.

Almulihi et al. [25] introduce a deep stacking ensemble model for early heart disease prediction, combining CNN-LSTM and CNN-GRU hybrid models with SVM as the meta-learner. The proposed model achieves superior performance over traditional machine learning models, but the study does not explore the model's robustness with real-world, unbalanced data or its computational efficiency.

Rath et al. [26] present an improved heart disease detection model based on ECG signals using an ensemble of deep learning models, including autoencoders (AE) and self-organizing maps (SOM). The ensemble model achieved accuracy and F1-scores above 97%, demonstrating the power of combining deep learning techniques for CVD prediction. However, the study does not discuss the scalability of the model for large datasets or its real-time application in clinical practice.

Bhan, Mangipudi, & Goyal [27] use deep and transfer learning-based methods to segment the left ventricle (LV) from cardiac MRI images for CVD diagnosis. The study compares CNN-based U-Net, VGG16, and ResNet152 models, with U-Net outperforming the other methods for LV segmentation. The research highlights the importance of LV position in diagnosing CVDs, though it does not address the limitations of the segmentation methods in noisy or incomplete MRI scans.

Nanni et al. [28] compare transfer learning and conventional machine learning models for the early diagnosis and prognosis of Alzheimer's disease (AD) using structural brain MRI. Their ensemble transfer learning approach achieved 90.2% AUC for distinguishing AD from cognitively normal subjects, showing the potential of transfer learning for

neuroimaging. However, the study does not evaluate the deep learning models on smaller datasets or assess their scalability for real-world applications.

Ismail et al. [29] review the application of transfer learning (TL) in improving electrocardiogram (ECG) diagnosis for cardiovascular diseases. The paper reviews 43 studies, identifying common pre-trained models like AlexNet, VGG, and ResNet, and discusses the impact of TL on ECG classification accuracy. While the study provides valuable insights into the use of TL models, it does not compare the performance of TL models with traditional machine learning techniques across various datasets.

Li et al. [30] explore the use of transfer learning for coronary heart disease classification using a small-sample medical imaging dataset. The study examines various deep learning models and evaluates their accuracy and loss metrics across several training epochs. While the research offers detailed insights into the models' performance, it does not fully evaluate the models' generalization performance across diverse, larger datasets or clinical environments.

Table 2.1: Summary of Literature Reviewed.

Author(s)	Model Used	Results	Contributions
Gulhane, M., & Kumar, S. [1]	ResNet50	Validation accuracy: 91.88%, improved to 94.27%	Proposed a deep learning model for heart disease detection from ECG images using ResNet50.
Maqbool, K., Singh, R. P., & Mehra, M. [2]	CNN + ResNet50	99.12% accuracy in classifying CVDs	Introduced CNN and ResNet50 for detecting arrhythmia, MI, and PMI from ECG images.
Mahmoud, S., Gaber, M., Farouk, G., & Keshk, A. [3]	Modified ResNet50	4.83% increase in accuracy from original	Modified ResNet50 for CVD prediction with improved accuracy compared to LeNet-5.
Xu, J., Chen, B., Liu, W., Dong, W., Zhuang, Y., Zhang, P., & He, K. [4]	Fine-tuned ResNet50	AUROC: 0.996 for DCM, 0.980 for HCM	Developed a fine-tuned ResNet50 for identifying HCM and DCM from ECG images with high predictive accuracy.
Yu, T., & Chen, K. [5]	Hybrid ML + Medical Imaging	96% diagnostic accuracy	Proposed a hybrid method combining machine learning with medical imaging modalities for CVD detection.

Anand, R., Lakshmi, S. V., Pandey, D., & Pandey, B. K. [6]	ResNet-50	98.14% accuracy in arrhythmia detection	Enhanced arrhythmia detection with ResNet-50, showing superior performance to conventional methods.
Mohanapriya, D., Mathivanan, P., Tharuneshwar, S., Somanathan, V., & Raaghul, K. [7]	DeepLabv3 + ResNet	Improved segmentation accuracy for myocardial abnormalities	Proposed a deep learning-based approach using DeepLabv3 for segmentation and ResNet for feature extraction in myocardial abnormality detection.
Hemalatha, S., Kavitha, T., Sivabalaselvamani, D., Naveena, N., & Haripriya, M. [8]	Capsule Networks + VGG16	VGG16: 99.8% accuracy	Compared CapsNet and VGG16 for early cardiac disease prediction, with VGG16 outperforming CapsNet.
Raju, K. S., Sekhar, K. C., Varma, M. K. S., Raju, K. L., Subrahmanyam, C. S., & Subbaraju, P. [9]	VGG16 + Federated Learning	Improved heart attack prediction accuracy	Integrated federated learning with VGG16 for privacy-preserving heart disease prediction from ECG signals.
Addisu, E. G., Yirga, T. G., Yirga, H. G., & Yehuala, A. D. [10]	VGG16 + Hybrid ML	92% accuracy for heart disease detection	Proposed a hybrid approach combining VGG16 and machine learning classifiers for heart disease prediction with explainability via SHAP values.
Van-Binh, N., & Thai-Nghe, N. [11]	VGG16 + Transfer Learning	Accuracy: 0.95, 0.96, 0.70 in three scenarios	Proposed using transfer learning with VGG16 for detecting cardiomegaly and aortic aneurysm from X-ray images.
Boulares, M., Alafif, T., & Barnawi, A. [12]	Transfer Learning (PCG signals)	No specific results; Benchmark proposed	Established a transfer learning benchmark for CVD recognition

			from phonocardiogram (PCG) signals.
Krishnan, V. G., Saradhi, M. V., Kumar, S. S., Dhanalakshmi, G., Pushpa, P., & Vijayaraja, V. [13]	DenseNet + Hybrid Optimization	Improved classification precision with smaller feature sets	Combined Lion Algorithm and Butterfly Optimization for feature selection, using DenseNet for heart disease prediction.
Mandava, M. [14]	Modified DenseNet201 + IDRSNet	Accuracy: 99.12%	Developed a hybrid system combining DenseNet201 and IDRSNet for more accurate CVD prediction.
Sivanathbabu, R., & Kamalakkannan, S. [15]	DenseNet	Accuracy: 91.86%	Proposed a DenseNet model for heart disease prediction from medical health records, outperforming existing models.
Aborokbah, M. [16]	EfficientNet-B0 + U-Net	Accuracy: 98.12%	Combined U-Net and EfficientNet for Alzheimer's disease MRI classification with high accuracy.

Table 2.1 provides a consolidated overview of the key research studies reviewed in this work, highlighting their objectives, methodologies, datasets, and major findings related to cardiovascular disease detection using deep learning. This summary enables a clear comparison of existing approaches and helps identify common trends, strengths, and limitations within the literature. By systematically organizing prior studies, the table establishes the research gap addressed by the proposed ensemble-based framework and justifies the selection of models, data types, and evaluation strategies adopted in this thesis.

2.2.1 Similar Applications

Multiple recent papers have used the methods of deep learning and transfer learning to detect cardiovascular disease, with promising results with a variety of imaging modalities and signal-based samples (See Table 2.2). ResNet50 and additional pretrained CNN models have been applied by many researchers to classify ECG images,

identify various conditions that include arrhythmia, myocardial infarction, and cardiomyopathy with high accuracy [1], [2], [4], [6]. Hybrid methods, combining a segmentation network, such as DeepLabv3 or U-Net, and feature extractor, such as ResNet or DenseNet, to detect structural abnormalities in cardiac MRI scans have been investigated by others [7], [27]. Another example is VGG16 and EfficientNet to detect disease in X-ray, echocardiogram and fundus images, which is an important idea as transfer learning is effective in a wide set of cardiac-related tasks [8], [11], [17], [21]. Ensemble learning is also not left behind, and there are studies on combining CNN, LSTM, GRU and others to enhance the stability and performance of predictions, especially in unbalanced or noisy data [24], [25], [26]. These applications demonstrate the increased attention to the idea of deep learning-based detection of cardiovascular diseases and justify the approach to the methodology of this thesis, comprising multiple pretrained CNNs and strategies of ensembles to enhance the validity of MRI-based classification.

Table 2.2: Summary of Similar Applications.

Author	Model Used	Accuracy	Contribution
Gulhane & Kumar [1]	ResNet50	94.27%	CVD detection from ECG images; improved accuracy with extended training.
Maqbool et al. [2]	CNN + ResNet50	99.12%	Classified normal/abnormal ECG patterns with high accuracy.
Xu et al. [4]	Fine-tuned ResNet50	AUROC: 0.996 (DCM), 0.980 (HCM)	Accurate cardiomyopathy detection from ECG images.
Yu & Chen [5]	Multi-modal ML + Imaging	96%	Combined MRI, echo, and X-ray for improved cardiac diagnosis.
Anand et al. [6]	Enhanced ResNet50	98.14%	Improved arrhythmia detection from ECG signals.
Hemalatha et al. [8]	VGG16 vs CapsNet	99.8%	Showed VGG16 is superior for early cardiac disease detection.
Addisu et al. [10]	VGG16 + ML Classifiers	92%	Hybrid VGG16-Random Forest improved

			interpretability with SHAP.
Mandava [14]	MDensNet201-IDRSRNet	99.12%	Hybrid DenseNet system for efficient CVD prediction.
Sivanathbabu & Kamalakkannan [15]	Optimized DenseNet	91.86%	Heart disease prediction from health records.
Aborokbah [16]	EfficientNet-B0	98.12%	Alzheimer MRI classification using EfficientNet.
Mahtab et al. [17]	EfficientNetB3	94.53%	Enhanced heart disease detection from echocardiogram images.
Mirsharifi & Keyvanpour [18]	EfficientNet + GradCAM++	97.84%	Interpretable Alzheimer's detection via feature visualization.
Dhanalaxmi et al. [19]	MobileNetV3	99.82% (UCI), 99.73% (Framingham)	Lightweight model for CVD classification.
Priyadharshini & Madheswaran [20]	CVD-TransNet	99.62%	Transfer learning ECG classifier outperforming CNN/SVM.
Sayin et al. [22]	InceptionV3	93.27%	MI detection from ECG images.
Baccouche et al. [24]	CNN + BiLSTM + BiGRU Ensemble	91–96%	Ensemble method addressing class imbalance.
Rath et al. [26]	AE + SOM + DL Ensemble	>97%	Strong ECG-based CVD prediction using ensemble models.

Table 2.2 presents a comparative overview of existing applications and systems developed for cardiovascular disease detection and related medical image analysis tasks. The table summarizes key characteristics such as application objectives, underlying technologies, input data types, and primary functionalities. This comparison helps highlight the practical capabilities and limitations of current solutions, emphasizing gaps in accuracy, interpretability, or deployment readiness. By reviewing similar applications, this table reinforces the motivation for developing the proposed deep learning–based ensemble system with enhanced performance, explainability, and mobile application support.

2.3 Gap Analysis

The literature analysis reveals that deep learning has achieved significant developments in the detection of heart diseases, but there are multiple significant gaps that prevent the application in practice and inspire the approach of this dissertation (See Table 2.3). The majority of works single CNN architectures, including ResNet50, VGG16, DenseNet, or EfficientNet, are used to classify ECG pictures, X-rays, or MRI images, and they tend to be highly accurate when learning on controlled datasets [1], [2], [8], [11], [17]. Nevertheless, such single-model systems are susceptible to overfitting and generalization to other patient groups, imaging disease states, or noisey settings does not necessarily have good results. The works by Gulhane & Kumar [1], Mahmoud et al. [3], and Anand et al. [6] show good outcomes, yet, they fail to estimate the strength of their models in a variety of datasets, or on the performance in the conditions of real clinical errors. In the same way, numerous MRI-centered experiments either deal with segmentation activities or test a small set of diagnostic groups, which still leaves the room to more comprehensive MRI-based disease classification schemes [7], [27]. This implies that there is a definite requirement of models that can learn more discriminatory and wider characteristics using MRI images.

The other significant lapse is that there is a shortage of ensemble learning techniques in the detection of cardiovascular diseases. Despite their potential to be very promising as they enabled better prediction consistency and class imbalance [24], [25], [26], numerous studies reviewed still use one architecture or one fusion method. Not many publications thoroughly discuss the possibility to use various pretrained CNN models to obtain supplementary information and minimize the weaknesses of each model. In addition, the approaches based on hybrid feature extraction or optimization, including those suggested by Krishnan et al. [13] and Mandava [14] do not test their generalizability to different imaging data sets, or compare their approaches to several deep learning baselines. This leaves room to ensemble based systems that formulate combinations of a variety of feature representations to enhance overall performance.

Lastly, interpretability and validation in the real case are poorly developed in most studies. Even though certain studies use visualization, like Grad-CAM [10], [18], the vast majority of them omit the thorough qualitative examination of predictions or illustrate the performance of the models in relation to unseen clinical samples. Most also do not have available feature-space visualization, able to confirm that a model indeed learns significant differences between classes of diseases. Thus, to deal with these gaps, a systematic workflow comprising of preprocessing, augmentation, numerous CNN

architectures, ensemble fusion, robust evaluation metrics and visualization methods are required. The proposed thesis seeks to address these issues by creating an ensemble system (using CNN) to detect cardiovascular disease using MRI on a human body, with better accuracy, enhanced generalization, and enhanced model interpretability.

Table 2.3: Summary of Research Gaps.

Identified Gap	Studies Showing the Gap	How This Study Addresses the Gap
1. Over-reliance on single CNN architectures with limited generalization across datasets.	Gulhane & Kumar [1], Maqbool et al. [2], Mahmoud et al. [3], Anand et al. [6]	Uses four different transfer-learning CNNs (VGG16, ResNet50, EfficientNet, DenseNet) to capture diverse features and reduce single-model limitations.
2. Limited research using MRI images for classification (most focus on ECG or segmentation).	Mohanapriya et al. [7], Bhan et al. [27]	Focuses directly on MRI-based disease classification, applying a full classification workflow instead of segmentation-only tasks.
3. Ensemble learning underused despite evidence that ensembles improve robustness.	Baccouche et al. [24], Almulihi et al. [25], Rath et al. [26]	Introduces three ensemble strategies—simple average, weighted average, and majority voting—to enhance accuracy and stability.
4. Lack of interpretability and limited real-case prediction analysis.	Addisu et al. [10], Mirsharifi & Keyvanpour [18]	Includes prediction visualization, Grad-CAM, and confusion matrices to improve interpretability and practical validation.
5. Few studies evaluate model performance under comprehensive evaluation pipelines.	Xu et al. [4], Van-Binh & Thai-Nghe [11]	Implements a full evaluation workflow: augmentation, individual model testing, ensemble comparison, ROC curves, and performance metrics.

Table 2.3 highlights the key research gaps identified from the reviewed literature and existing applications in the domain of cardiovascular disease detection. The table summarizes limitations related to dataset usage, model performance, lack of ensemble strategies, insufficient interpretability, and limited real-world deployment. By clearly outlining these gaps, the table establishes the need for a more robust, accurate, and explainable deep learning framework. These identified gaps directly motivate the proposed CNN-based ensemble approach with Grad-CAM visualization and mobile

application deployment, positioning this study as a meaningful advancement over existing work.

2.4 Summary

This chapter gave the background information and review of the current literature that applies to cardiovascular disease detection with the help of the deep learning method. It presented the application of different models of the ResNet50, VGG16, DenseNet, EfficientNet and ensemble framework to ECG, X ray, MRI, and other medical imaging modalities with high results, however, showing limitations in generalization, robustness, and interpretability. Based on the research conducted on these studies, several gaps in the research were observed, which comprised of the excessive use of single CNN models, little MRI based classification study, and the underutilization of ensemble learning, which has been verified to be beneficial. These lessons show that a more effective and valid method of detecting CVD is necessary. This chapter ended by connecting the existence of these gaps to the rationale of developing this thesis to formulate a proper justification of creating CNN-based ensemble system capable of combining various feature representations and offer better diagnostic results based on cardiac MRI images.

Chapter 3

Research Methodology

This chapter presents the complete methodology for developing the proposed cardiovascular disease detection system using a deep learning–based ensemble approach. It begins with a detailed description of the cardiac MRI dataset, followed by the preprocessing, data augmentation, and dataset splitting strategies applied to prepare the images for model training. The chapter then explains the architecture and working principles of the four-transfer learning–based CNN models—ResNet50, VGG16, DenseNet121, and EfficientNetB0—used for image-level classification, along with the ensemble learning techniques designed to enhance overall classification performance. Model evaluation procedures and Grad-CAM–based visualizations are also discussed to ensure interpretability and transparency of predictions. Finally, the methodology outlines the integration of the optimized ensemble model into a mobile application, highlighting its practical applicability. Overall, the chapter clearly defines each stage of the workflow, ensuring a structured, reproducible, and technically rigorous research framework.

3.1 Methodology

3.1.1 Overview

This paper proposes a deep learning-based framework of automated cardiovascular disease detection in cardiac MRI images in accordance with a systematic and reproducible methodological pipeline. The process starts with the acquisition and preprocessing of the dataset systematically: resizing, normalization, denoising, and contrast enhancement have to be conducted to maintain the quality of the image. In order to counterbalance the imbalance between the classes data augmentation is used before dividing the dataset into the training, validation and test sets. A series of already trained convolutional neural network architectures, which include ResNet50, VGG16, DenseNet121 and EfficientNetB0, are then used with the assistance of transfer learning to extract robust features and classify them. The models are considered separately and then they are collectively implemented into an ensemble learning plan incorporating voting, average, and weighted method to improve the predictive accuracy. Lastly, the visual explainability of model choices is offered with the help of Grad-CAM, and the fine-

tuned ensemble model is ready to be introduced into a mobile app, which shows the high practical relevance of the suggested system.

3.1.2 Proposed Methodology

Figure 3.1 shows the general methodological framework used in this study in the detection of cardiovascular disease based on deep learning-based ensemble models. The suggested strategy is a step-by-step pipeline with the initial phase based on the purchase of cardiac MRI images and the further phases of systematic data preparation, model development, evaluation, and deployment. All parts of the workflow should be developed to be able to guarantee data quality, model soundness, and clinical relevance, and also be reproducible and understandable. The flowchart diagrammatically summarizes the interrelations between individual processing and learning building blocks in creating a complete end-to-end diagnostic system.

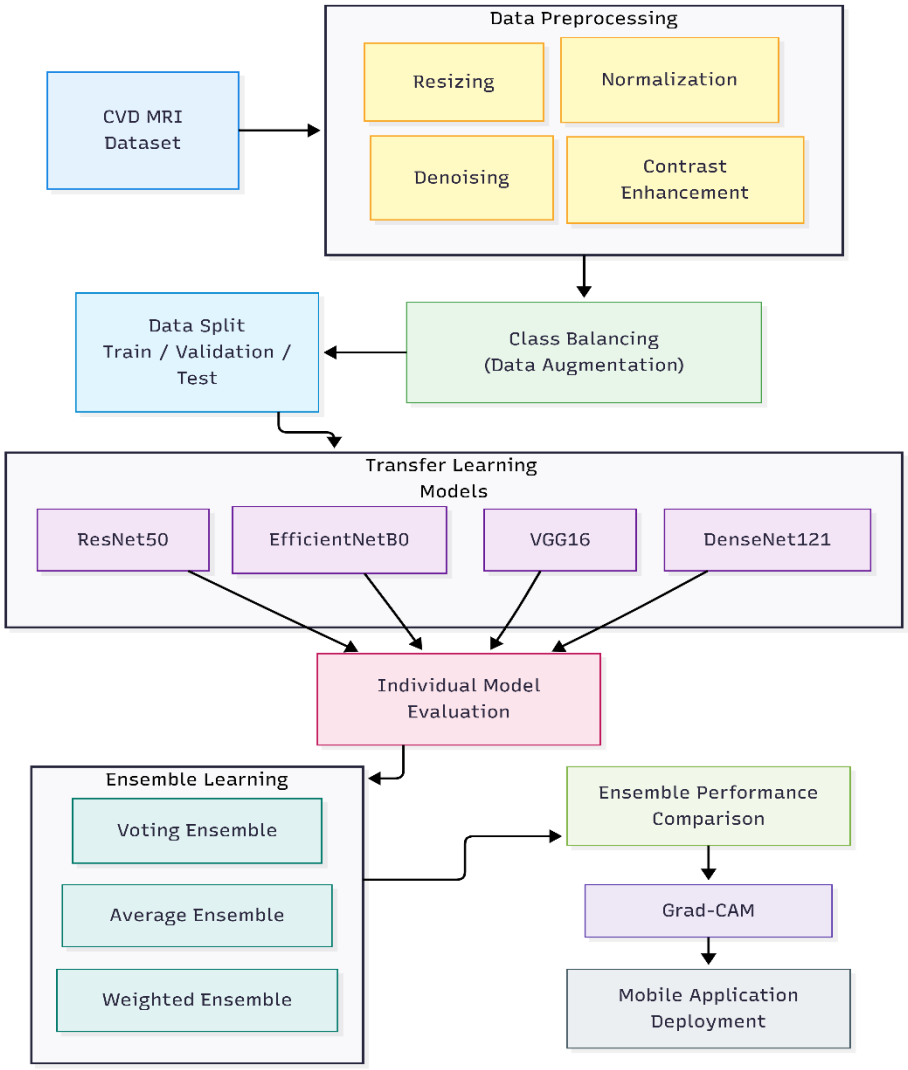


Figure 3.1: The Methodological Flowchart

The cardiac MRI dataset will be processed in multiple stages (preprocessing, denoising, contrast and resizing) so that it can first be standardized (image inputs represented by resizing and normalization) to ensure more visible features (Figure 3.1). Class imbalance is then mitigated first by applying data augmentation after which the dataset is separated into training, validation and test sets. To apply transfer learning, several CNNs are used, namely, ResNet50, VGG16, DenseNet121, and EfficientNetB0, to retrieve the deep features and classify them, after which each model is evaluated separately. The performances of these models are added together through ensemble learning approaches like voting, average and weighted methods to enhance overall performance of prediction. Lastly, Grad-CAM is applied to give visual insights into the model choices and the model of the optimized ensemble is implemented in a mobile application, which makes it possible to have cardiovascular disease detection in practice and understanding.

3.1.3 Functional and Nonfunctional Requirements

Cardiovascular disease detection system functionalities are supposed to be effective, reliable and practical so as to be applicable to the study participants; to ensure the same, following non-functional and functional requirements are based on the objective and methodology of the study:

Functional Requirements:

- CVD Image Classification: The system will have to properly classify cardiac MRI images into the correct cardiovascular disease-related categories according to learned deep features.
- Data Preprocessing Capability: The framework should automatically scale the input images and normalize, denoise as well as optimize contrast to standardize the input and enhance the quality of the features before model training.
- Class Balancing Mechanism: The mechanism should use data augmentation methodologies to reduce the class imbalance and enable balanced and unbiased models learning.
- Model Training and Transfer Learning: The system should be able to train and fine-tune various pre-trained CNN architectures, such as ResNet50, VGG16, DenseNet121, or EfficientNetB0, and use them with successful feature extracting and classification.
- Individual Model Analysis: The framework should also analyze each of the trained models separately in terms of the standard performance metrics to test

the diagnostic ability of each model.

- Ensemble Learning Support: The system should be able to collectively predict the individual models based on the voting, averaging, and weighted ensemble techniques to enhance the aggregate detection accuracy and strength.
- Model Interpretability: To help facilitate clinical interpretability, the system should be able to produce Grad-CAM images, which are the visually important areas of the image affecting model predictions.
- Mobile Application Deployment: The completed ensemble model should be deployable on a mobile application environment so that it can be used in a real-world situation and be accessible.

Nonfunctional Requirements:

- Reliability: To prevent overfitting and augmentation, regularization and ensemble, the model will not provide alternate predictions on fresh test batches.
- Scalability: The model must be capable of handling larger data sets, new MRI classes or new CNN models without requiring any major changes to the model.
- Performance Efficiency: The system should provide reasonable training and inference times by means of the use of the GPU-conformable environments and simplified deep learning piping.
- Usability: The processes should be easy to follow, and the code is structured under clean code and the documentation is clear and the visualization of the outputs of the code can be interpreted by both the researcher and clinicians.
- Maintainability: All scripts and notebook files should be mutable and version controlled and must be organized in a manner that they are easy to modify to enable more exploration or improve the system.
- Ethical Compliance: The system must be ethical and therefore assure the MRI data, and make predictions to support, but not to supersede the clinical decision-making.

3.1.4 Data Flow Diagram

The Level 1 Data Flow Diagram (DFD) of the proposed cardiovascular disease detection system is shown in Figure 3.2, it shows the movement of data through the key functional processes of the system. The diagram is interested in converting cardiac MRI images input data into final diagnostic output data and emphasizes interactions of data sources, processing modules, and system outputs. It gives a concise conceptual picture of the working of the internal components to aid in automated disease detection and decision-

making.

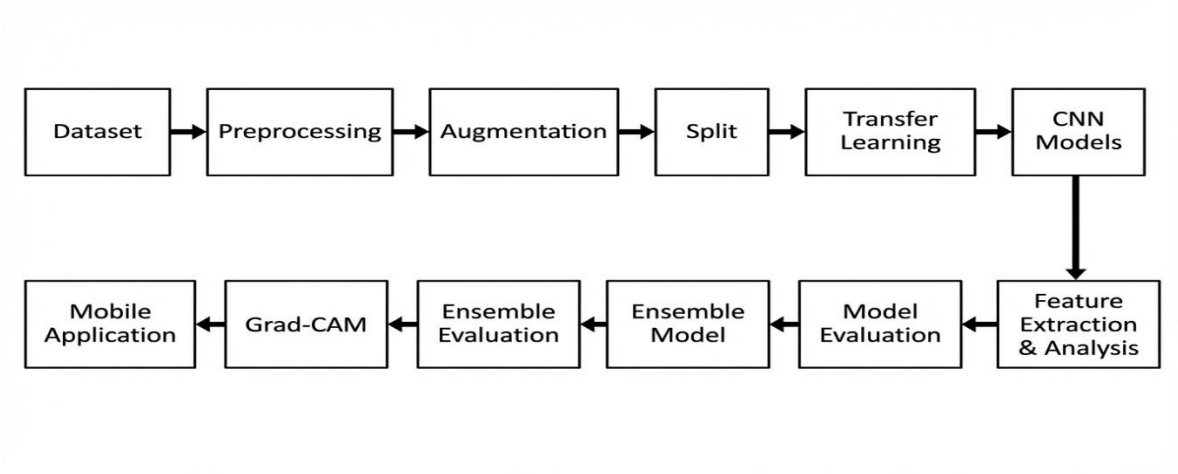


Figure 3.2: Proposed Data Flow Diagram

In Figure 3.2, the data preprocessing module receives the cardiac MRI images, on which the data processing process is carried out. The data processing process consists of resizing, normalizing, denoising, and contrast enhancement followed by enhancing the quality of the data. The processed images are afterwards subjected to the process class balancing and data splitting which makes sure that the classes are fairly represented and that there is a proper division into training, validation, and test sets. The ready data are inputted into various transfer learning-based CNNs to extract and classify features, and it is then succeeded by separate and combined-level prediction. In view of this, the system generates interpretable outputs in the form of performance results and Grad-CAM visualizations which are sent to the deployment module to be incorporated into the mobile application at the end of the end-to-end data movement of the system.

3.1.5 UI Design

The user interface (UI) of the suggested CAD detection system is highly simplified and clear and easy to use, so that the medical personnel and the non-expert users of the system can easily interact with it. The main interface has a clean design where a specific section, Cardiac MRI Analysis, has been created where the user is able to upload cardiac MRI images in a drag and drop format or browse the device. There are also clear guidelines like the supported formats (PNG, JPG) and size restrictions so that the user can easily follow the steps of uploading the file. The design uses a simple color scheme and organized card elements to look and feel professional in the field of medicine but increase legibility. Icons and graphical indications have been selected in order to create user confidence and alleviate cognitive burden. In general, the UI design is focused on the priorities of accessibility, responsiveness, and an intuitive interaction, creating a

user-friendly gateway to the implementation of the deep learning-based cardiovascular disease detection.

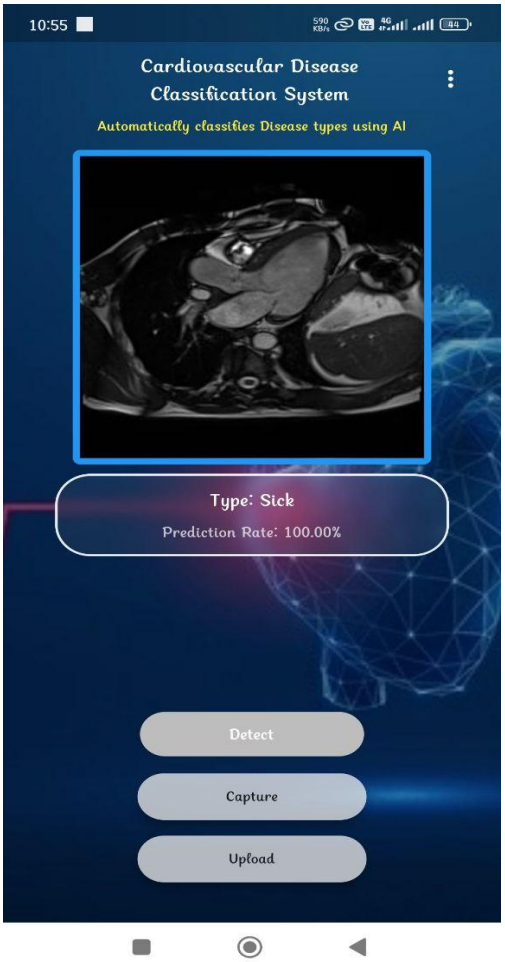


Figure 3.3: The proposed UI design for the Mobile Application

3.2 Detailed Methodology and Design

3.2.1 Dataset

The sample in this research comprises of 63,648 multiparametric cardiac magnetic resonance (CMR) images of 1,224 patients which are publicly available in an open-access repository. Out of these patients, 722 will be healthy individuals, and 502 will be diagnosed with coronary artery disease (CAD), which is a clear-cut binary classification environment. The dataset has 13 image slices of the patient where four different types of CMR sequences have been performed (Late Gadolinium Enhancement (LGE), Perfusion imaging, T2-weighted imaging and Steady-State Free Precession (SSFP)). The pictures are taken of long-axis (two-chamber, three, and four chambers) and short-axis,

the latter being the ones between the heart base and the apex. Every slice will be acquired at two jump angles, which will provide the overall representation of the cardiac anatomy and physiological features. This dataset will be divided into two image-level categories, which are CAD-positive and healthy without processing the information at the patient-level. The invasive coronary angiography, which is regarded the clinical gold standard, was used as the basis of the CAD labels and guaranteed the high level of diagnostic reliability. On the whole, the data set consists of 26,104 positive images of CAD and 37,544 healthy images, thus, it is appropriate to train and test the deep learning models to detect cardiovascular disease based on images. Overall, this data provides high-quality, medically-proven MRI images featuring rich diagnostic properties and is therefore suitable in the training and testing of the proposed cardiovascular disease detection system (see Table 3.1).

Table 3.1: Dataset Specifications

Properties	**Values**
Image Resolution	224 × 224 pixels
Format	JPEG (as provided on Kaggle)
Total Images	~63,000 images
Classes	2 classes: Normal, Sick

The sample data of this study will consist of two classes, and each of them will have a different heart health disorder which will be registered through MRI examination. These classes form the binary classification task and form the deep learning models.

- Normal: These are MRI images of individuals who did not have a cardiovascular disease. The images are normally described by the presence of healthy myocardial tissue, normal chamber morphology and similarity of contrasts of different MRI sequences. These samples facilitate the model in gaining a knowledge of base lines of heart structures and normal functioning features.
- Sick: This kind of images contains MRI images of a patient who has been diagnosed to have a coronary artery disease (CAD). The deformities that the images can exhibit are: reduced perfusion, tissue injury that is highlighted in LGE scans besides abnormal and uncharacteristic structures that are seen in SSFP and T2-weighted images. These samples are significant in teaching the model, how the pathological changes will be revealed in MRI scan.

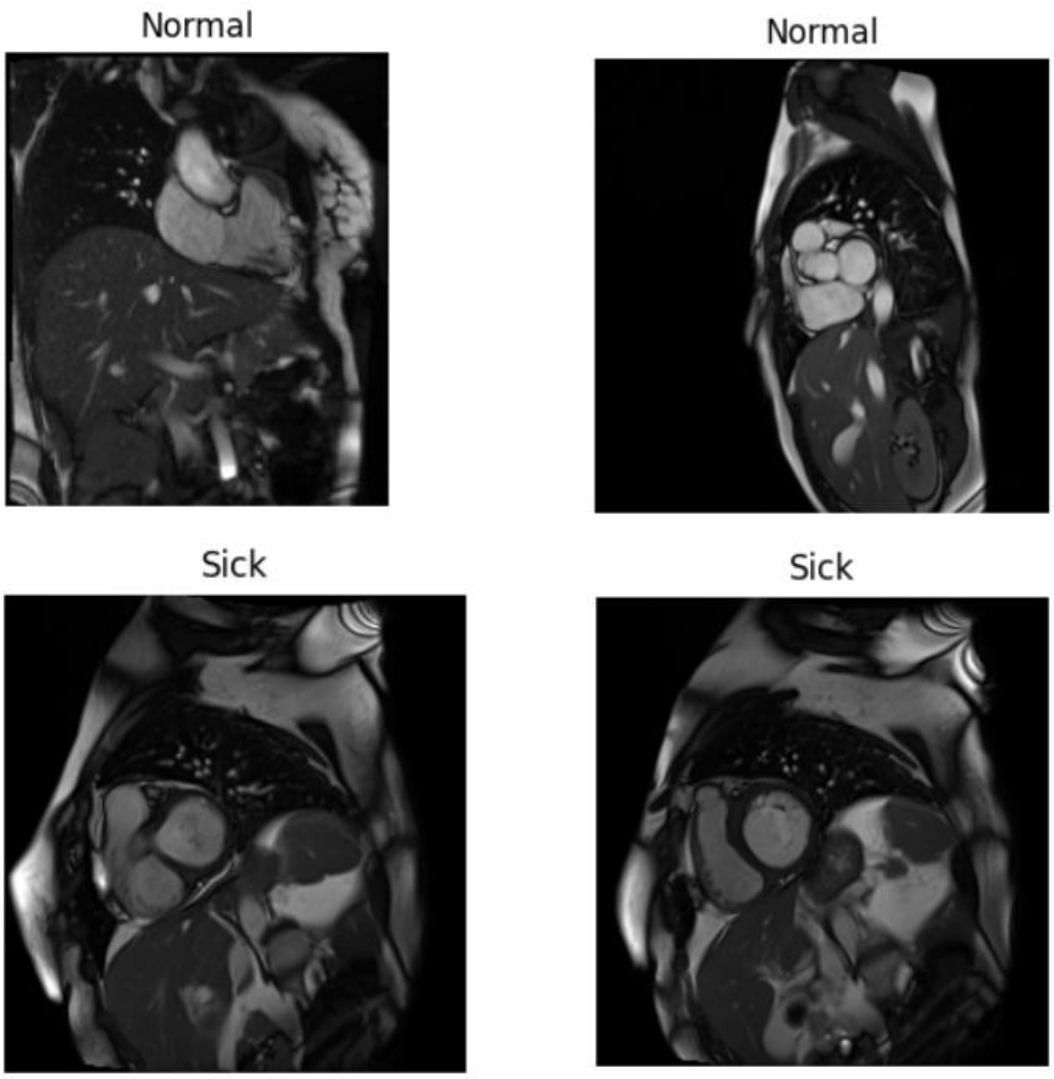


Figure 3.4: Sample Image from Each Class

3.3.2 Preprocessing Steps

Preprocessing phase aims at ensuring that all MRI images are standardized, cleaned, and ready to be used to train a deep learning model. The procedures that were followed are:

- Image Resizing: All images were down-size to 224 x 224 pixels to adjust to the input of the pretrained CNN models. This ensures standardization as well as reduced work in computing.

$$I_{resized} = Resized(I, W_{target}, H_{target})$$

- Normalization: The pixel values have been normalized to the 0-1 scale, and it helped to stabilize the training process and improve the flow of the gradient and reduce the

numerical difference between the images.

$$I_{norm} = \frac{I - I_{min}}{I_{max} - I_{min}}$$

- Dataset Splitting: It was divided into 70 training, 15 validation and 15 testing data. It can be used to aid in a fair assessment and remove the leakage of data but not the balance of classes.
- Data Augmentation: The training images were transformed by means of rotations, flips, zooms and shifts. It makes the datasets more varied, reduces the overfitting, and encourages generalization.

$$I_{aug} = T(I)$$

- Label Encoding: The two types of the dataset, that is the normal and the Sick was coded into a numerical (0 and 1) data, which allowed the deep learning models to work with them.

$$y_{encoded} = Encode(y)$$

- Denoising: Noise that is generated with the acquisition of the MRI is minimized to increase image quality and remove non productive changes. This will assist the model to concentrate on the significant anatomical features instead of the chance occurrences.

$$I_{denoised} = I - N$$

- Contrast Enhancement: Contrast enhancement The enhancement of cardiac regions of interest by raising the contrast between low-intensity and high-intensity regions. This increases the conspicuousness of small structural features to extract features.

$$I_{enhanced} = C(I)$$

- Input Standardization for CNNs: Feed aheads are standardized to the input capabilities of the desired CNN architectures in both arrangement of channels and constant shape. This also guarantees smooth integration with transfer learning models.

$$X_{input} = Format(X)$$

3.2.3 Model Description

This research capitalizes on the well-defined set of state-of-the-art convolutional neural network (CNN)-based cardiovascular disease detection models ResNet50, VGG16,

DenseNet121, and EfficientNetB0 to identify state-of-the-art and interpretable cardiovascular disease detection using cardiac MRI images. Both models are efficiently chosen based on their individual architectural optimism and their recognition in achieving an excellent processing in deep visual representation learning. The entire programming cycle of the suggested models that is in line with the code files adopted are detailed as follows.

- Input Layer Initialization: All CNN models are fed with equalized images of cardiac MRI having equal dimensions and channels. This guarantees flow of information and enables all the architectures to run on the same conditions of the experiment.

$$X_{input} = Input(I_{preprocessed})$$

- Superior CNN Backbone Initialization (Transfer Learning): The pre-trained weights of every CNN architecture are used to initialize the models, which allows the models to inherit rich and hierarchical visual knowledge.
    - ResNet50 has the advantage of residual connections, which reduce the problem of vanishing gradients and allow deep and stable learning.
    - VGG16 offers a clean and sequential design that is more efficient and allows observing fine spatial patterns with a very fine precision.
    - DenseNet121 vendors feature reuse by connection with high density which enhances gradient flow and learning performance.
    - EfficientNetB0 is able to obtain impressive performance with a smaller number of parameters through an optimal choice of network depth, width, and resolution.

$$W_{init} = W_{pretrained}$$

- Strategic Layer Freezing and Fine-Tuning: To achieve to the maximum the generalization of learned representations and, at the same time, retain learned representations, the early convolutional layers are frozen, and deeper convolutional layers are fine-tuned selectively. This enables the models to learn high-level representations to patterns unique to cardiac MRI without being unable to retain the underlying visual knowledge.

$$W = \{W_{frozen}, W_{trainable}\}$$

- Global Average Pooling for Feature Compression: The obtained features maps are converted into small feature vectors by global average pooling. The step improves the computational efficiency and prioritizes the most salient activations associated with

the patterns of disease.

$$f_{gap} = \frac{1}{N} \sum_{i=1}^{N} f_i$$

- Powerful Fully Connected Classification Head: Dense layers are trained to provide complicated nonlinear association between extracted representations and disease categories. Regularization of dropout is used to enhance generalization and guard against overfitting of the models.

$$z = W\, f_{gap} + b$$

- Probabilistic Output Layer: It is a sigmoid-activated output layer which gives the probability estimates of coronary artery disease, which can use to create confident and interpretable image-based prediction.

$$P(CAD) = \sigma(z)$$

- Independent Model Training and Performance Excellence: CNNs are trained separately, and this has enabled them to take full advantage of their architectural benefits. The variability of learning behavior among models makes the difference in terms of high individual performance and complementary forecasting patterns.
- Ensemble Intelligence and Decision Fusion: Predictions of all CNN models are combined by using average, weighted average and vote strategies. This system intelligence leverages the synergy of all architectures, and this power has a great enhancement in robustness, stability, and diagnostic accuracy.

$$P_{ensemble} = \frac{1}{M} \sum_{m=1}^{M} P_m$$

- Grad-CAM–Driven Explainability: Grad-CAM is used to create class-discriminative heatmaps that can visually show areas of the cardiac MRI images that contributed to the predictions. This interpretability process enhances trust in clinical clinicians and fills the medical decision-making and deep learning disconnect.

$$L_{GradCAM} = ReLU(\sum_{k} \alpha_k A_k)$$

- Final Prediction with Clinical Insight: The individual output coupled with Grad-CAM visual explanations provides a precise and clear image-level diagnosis. Such deep feature learning, ensemble intelligence coupled with explainable AI synergy creates a powerful and clinically significant cardiovascular disease detecting system.

3.3 Project Plan

The project plan is arranged to be tightly aligned with the suggested methodological flowchart and provides a logical flow between data preparation and model deployment. The timeline is made to cover systematically all the key phases of the research as well as provide enough time to experiment, evaluate, and document. Preliminary phases are devoted to the development of a powerful theoretical basis and the clear definition of the research process which directly predisposes the organization of the deep learning based ensemble framework.

Table 3.2: The Gantt Chart of Project Timeline

Process	May'25	June'25	July'25	Aug'25	Sep'25	Oct'25	Nov'25	Dec'05
Working Plan								
Theoretical Study								
Literature Review								
Data Collection & Data Preprocessing								
Model Design + Methodology Writing								
Ensemble Learning								
Grad-CAM Analysis & Mobile Deployment								
Report Writing								

The project is initiated by the working plan development, theoretical research, and comprehensive literature analysis to learn the current cardiovascular disease detection methods and the gaps in the research. This is accompanied by the acquisition and organization of the cardiac MRI images into datasets according to the classification strategy that has been stipulated. Preprocessing of data and balancing of classes are then done to make quality inputs to the development of the model. After that, several transfer learning models are developed, which is then trained, and tested one by one and finally incorporated into an ensemble learning model. Grad-CAM-based interpretability

analysis is performed on the result of ensemble analysis in order to give visualized explanation of what a model predicts. Last but not least, the study ends with the report writing and the preparation of the mobile application deployment, which will guarantee the scientific rigor of the proposed system and its practical applicability.

3.4 Task Allocation

The division of labor in this project was carried out in a manner that ensured smooth running of the project as the project was being developed to the last development processes till its final documentation. Sharing out of duties was based on the rational working process and technical requirement. The first stages were to determine the scope of the research, the theoretical research, and the literature review to develop the sound conceptual framework. This was followed by data collection, data exploration and preprocessing that consisted of resizing, normalization and augmentation of MRI images. The researcher created model designs, implemented and trained on transfer learning architectures, which included ResNet50, VGG16, DenseNet121 and EfficientNetB0, after which each and each ensemble model was evaluated individually and through ensembles. Grad-CAM interpretability analysis and mobile application implementation were then implemented to improve explainability and applicability in the real world. Report writing, graphical representation of results and polishing of records were assigned as the last stage in order to make sure that the thesis was good enough academically.

3.5 Summary

The chapter involved a detailed procedure of detecting cardiovascular diseases with the help of a CNN ensemble-based approach based on deep learning. It outlined the entire research process, data specifications, classifications, and preprocessing operations and model architecture to provide a well-structured and reproducible framework. Combination of various developed CNN structures and ensemble learning overcame the shortcomings of single models as it enhanced the predictive accuracy and stability. Also, Grad-CAM was used to enhance visual interpretability that enhanced transparency and clinical relevance. Lastly, the fact that the proposed system includes the deployment of a mobile application points to the practical application of the suggested algorithm, as it can be used in real-life scenarios to identify cardiovascular disease in real-time and with ease.

Chapter 4

Implementation and Results

This chapter shows how the proposed methodology was carried out and gives the performance of the developed models. It provides a description of training environment, hyperparameters and step wise implementation of every CNN architecture. Individual model and ensemble strategy results are provided with a detailed evaluation result, such as accuracy, precision, recall, F1-score, ROC curves and confusion matrices. The Grad-CAM examples are visualizations that explain the effectiveness of the models in separating the images of the Normal and Sick MRI. All models have also been compared in the chapter to find the most effective one and the reason why the ensemble method is able to generate more stable and reliable predictions. All in all, this chapter presents evidence of effectiveness of the system and its potential usefulness.

4.1 Environment Setup

Table 4.1 reflects the most agreed training parameters that were used in all experimented deep learning models to provide a fair and similar evaluation. The input image size was set to 224x 224x3 to ensure that it aligned with the trained CNN drugs and because the spatial and channel dimension were uniform across all experimental results. Applying the same image resolution can also be used to minimize variability in computations and the performance variation is due to the model architecture and not the input setting.

Table 4.1: Common Parameters Used Across All Experimented Models

Parameter Name	Value Used
Image Size	224 × 224
Batch Size	128 (Training & Validation), 16 (Testing)
Epochs	25
Optimizer	Adamax
Learning Rate	0.001

Training and validation were performed with a batch size of 128 to strike a balance between good use of the GPUs and stable gradient updates and a smaller batch size was used in the testing phase, 16, because of the limit on memory availability and to provide

credible inferences. Each model was trained to 25 epochs, and this was enough to enable convergence, and not to overfit unnecessarily, in combination with early stopping methods. Adamax optimizer has been selected because it is stronger against the noisy gradient as well as sparse updates. The learning rate of 0.001 was used throughout the models to ensure a constant convergence behavior and optimization dynamics to help one make a meaningful comparison of the model performance when they are trained under the same conditions.

Table 4.2 shows the typical dataset separation plan that is used consistently across all the experimented models so that there is consistency and impartial evaluation of performances. The entire dataset was split into training, validation and test sets in a ratio of 70:15:15 ratio which is a common procedure in medical image analysis using deep learning. Such a divisiveness enables the models to learn efficiently with the help of a large enough training set with independent data being kept aside to provide an unbiased validation and the ultimate evaluation of performance.

Table 4.2: Common Data Split for All Experimented Models

Dataset	In Percentage	Number of Images
Training Set	70%	52,544
Validation Set	15%	11,260
Test Set	15%	11,260

The parameters of the models were optimized on the training set which included 70 percent of the total images and it learned discriminative patterns that defined normal and diseased cardiac conditions. The validation set with 15 percent of data was utilized in tuning the hyperparameters, and it was used to monitor the generalization of the model during the training process and help in avoiding overfitting. The rest 15 percent of images constituted the test set that was to be used strictly during the end evaluation, to determine the actual generalization performance of each of the models. Dividing the data in all experiments was maintained to make a fair comparison of the results and to be able to interpret the difference in performance of the models that were tested.

4.2 Performance Analysis

In evaluating the effectiveness of machine learning models for the study, appropriate performance metrics must be used to provide insights into model accuracy, reliability, and generalization capabilities. The following metrics are commonly used in classification tasks, especially in the context of agricultural disease detection:

4.2.1 Accuracy

The proportion of correctly classified instances (both positive and negative) to the total instances. Accuracy gives a quick overview of model performance but can be misleading in imbalanced datasets where one class significantly outnumbers the other.

$$Accuracy = \frac{TP+TN}{FP+FN+TP+TN} \quad (1)$$

4.2.2 Recall

The ratio of correctly predicted positive observations to all actual positives. Recall is particularly important in scenarios where a positive case (such as a diseased plant) could lead to severe consequences, like crop loss.

$$Recall = \frac{TP}{(FN+TP)} \quad (2)$$

4.2.3 Precision

The ratio of correctly predicted positive observations to the total predicted positives. Precision is crucial in applications where the cost of false positives is high. In this study, high precision indicates that when a disease is predicted, it is likely to be true.

$$Precision = \frac{TP}{(FP+TP)} \quad (3)$$

4.2.4 F1-Score

The harmonic meaning of precision and recall, providing a balance between the two metrics. The F1 score is especially useful when dealing with imbalanced classes, as it considers both false positives and false negatives, offering a more comprehensive view of model performance.

$$F1\ Score = 2 \times \frac{Precision+Recall}{(Precision \times Recall)} \quad (4)$$

4.2.5 Confusion Matrix

A table used to describe the performance of a classification model, showing the true vs. predicted classifications. The confusion matrix provides insights into the types of errors made by the model, allowing for more targeted improvements.

4.2.6 Receiver Operating Characteristic (ROC) Curve

A graphical representation of a classifier's performance across various threshold settings, plotting the true positive rate (recall) against the false positive rate. It illustrates the diagnostic ability of a binary classifier system as its discrimination threshold is varied. It plots the True Positive Rate (TPR), or recall or sensitivity, against the False Positive Rate (FPR) at various threshold settings.

Area Under Curve (AUC): The area under the ROC curve provides a single metric to assess model performance; a value closer to 1 indicates a better model.

In this study, the Receiver Operating Characteristic (ROC) curve and Area Under the

Curve (AUC) are crucial for evaluating classification models. They allow for visual comparisons of model performance across various thresholds, facilitating optimal threshold selection by balancing sensitivity and specificity. Additionally, the ROC curve is robust against class imbalances, providing reliable assessments where accuracy may mislead. However, ROC and AUC should complement other metrics like precision and recall for a comprehensive evaluation of model effectiveness.

4.3 Results and Discussion

4.3.1 Overview of Results

The following section provides a summary of the experimental findings after the proposed cardiovascular disease detection framework, according to the methodological pipeline outlined in Chapter 3. It equips itself with the performance of single-transfer learning-based CNNs; ResNet50, VGG16, DenseNet121, and EfficientNetB0, which is compared across parallel training configurations, data splits, and evaluation criteria to provide fair comparison. The findings indicate that there are significant discrepancies in performances of the models that show how each architecture is effective in the learning of discriminative features on cardiac MRI images. Moreover, the ensemble learning techniques are used in order to combine the best individual model strengths resulting in a high classification accuracy and robustness. The quantitative results are supplemented with the inclusion of Grad-CAM-based visualizations to present their interpretable results, and the last optimized ensemble model is considered to be deployed to mobile applications, and the experimental results are consistent with the practical goals of the study.

4.3.2 Training Performance Analysis

Figure 4.1 shows the training history comparison of all the experimented CNN models in terms of loss, accuracy, precision and recall after 25 training epochs. It can also be observed in the loss curves that DenseNet121 has the most stable and consistent convergence as both the training and validation loss curves fall smoothly and have a small gap thus showing a great deal of generalization and learning. VGG16 also shows gradual decrease of the loss, however with relatively higher ultimate figures than the one of DenseNet121 which indicates good and relatively poor optimization ability. ResNet50 has moderate convergence behavior with the loss decreasing slowly but higher than that of DenseNet121 and VGG16. Conversely, EfficientNetB0 has insignificant increase in loss throughout the epochs, which might imply that it does not have enough

learning capacity with the given set- up.

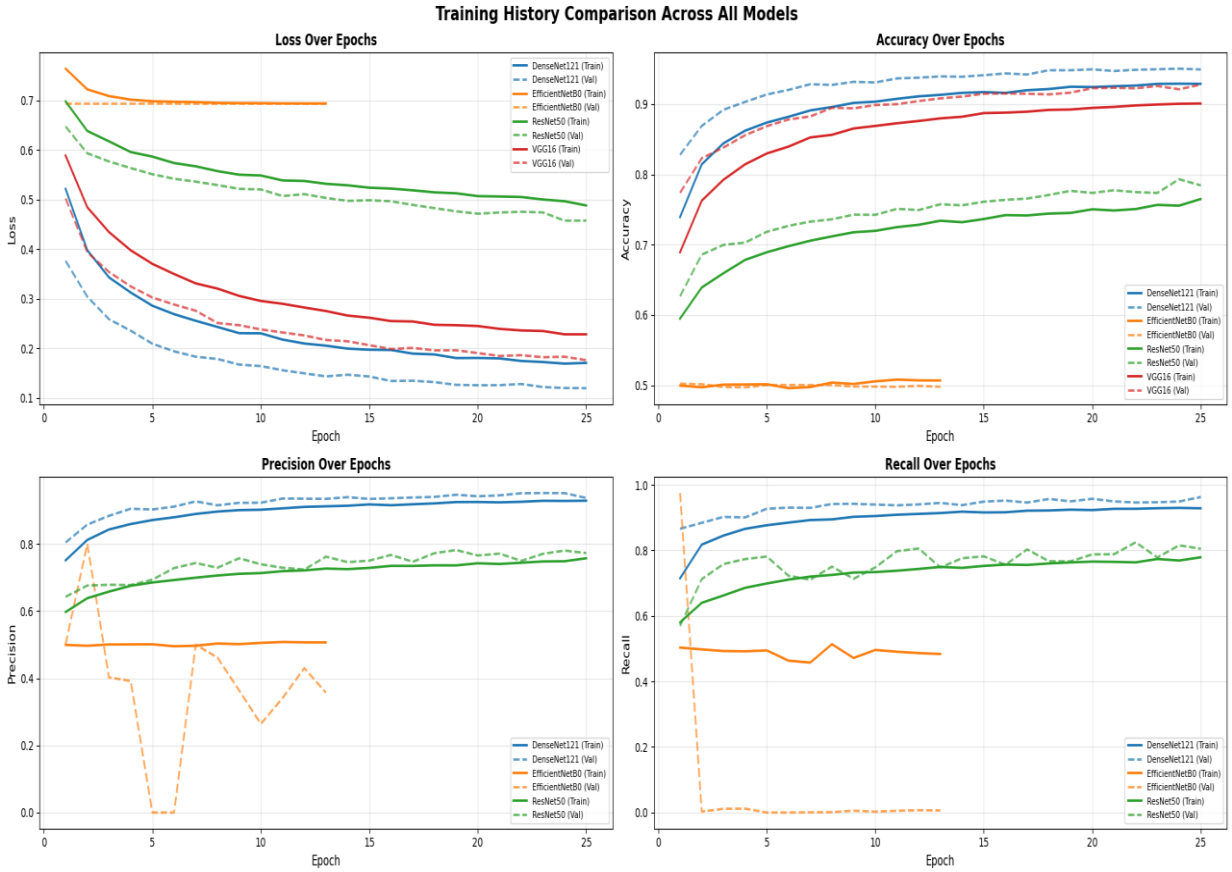


Figure 4.1: Training History Comparison Across All Models

These observations are further supported by the trends in the accuracy. DenseNet121 is the highest and most consistent at training and validation accuracy with a score above 94 percent and close curve alignment which shows minimal overfitting. The VGG16 has higher accuracy development and convergence rates with ResNet50 showing moderate rates of accuracy increase with a training to validation gap that is prominent but controlled. The performance of EfficientNetB0 at a stage of training does not exceed the range of chance, a factor which demonstrates that it cannot learn the discriminative characteristics of such a dataset effectively. The same trends can be seen in the precision and recall plots where DenseNet121 is always found to be showing high and balanced precision-recall rates, which are then succeeded by VGG16 and ResNet50. EfficientNetB0 demonstrates poor stability in precision and an extremely low recall, which proves that it is not sensitive to diseased cases. On the whole, it is evident that Figure 4.1 illustrates the advantage of DenseNet121 and VGG16 when they are used in the process of training, which is why they deserve their place in the suggested ensemble model.

4.3.3 Feature Distribution Analysis (t-SNE)

The t-SNE visualizations provide insight into how well each model separates Normal and Sick MRI features in a reduced two-dimensional space. VGG16 and DenseNet121 show clear and compact clusters, indicating that their feature representations are highly discriminative. ResNet50 demonstrates moderate separation, with some overlap between classes, suggesting less distinct feature learning. EfficientNetB0 shows the weakest separation, with considerable mixing of Normal and Sick points, which aligns with its lower classification performance. Overall, the t-SNE results confirm that models with stronger feature clustering—particularly VGG16 and DenseNet121—tend to achieve higher accuracy and robustness in downstream classification.

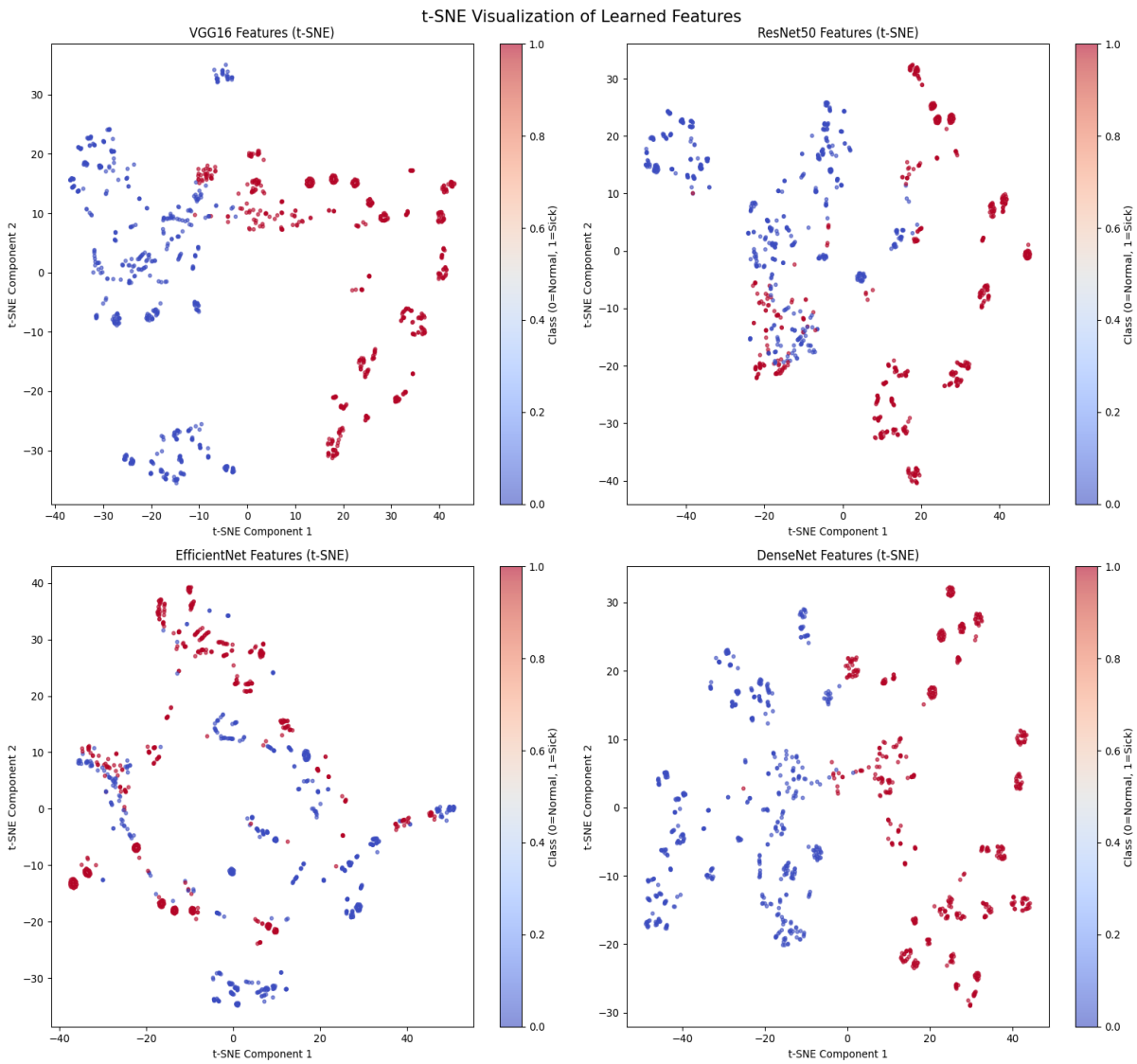


Figure 4.2: t-SNE Visualization of Learned Feature Representations

Figure 4.2 demonstrates the t-SNE projections of the feature embedding learned by

VGG16, ResNet50, EfficientNetB0, and DenseNet121 to show the effectiveness of each of the models to distinguish between the Normal and Sick classes of cardiac MRI in a two-dimensional space that is reduced. The most evident separation is between VGG16 and DenseNet121 that are represented by two distinct clusters with little overlap, which denotes features discrimination. ResNet50 shows recovery levels of separation, but the classes still have some intermixing. EfficientNetB0 has the lowest clustering, and significantly overlapping and scattered points, which indicates a poorer performance in classification. Comprehensively, the visualization supports the notion that VGG16 and DenseNet121 are better at extracting discriminative features, which are in line with their better quantitative performance.

4.3.4 Individual Model Evaluation on the Test Set

The performance of single CNN models on the test data in terms of individual classification with regard to class-wise precision, recall, F1-score, total accuracy, and AUC-ROC is outlined in Table 4.3. The findings prove that the models have distinct performance differences, and indicate the capabilities and weaknesses of each architecture as applied to cardiovascular disease detection with cardiac MRI. The table contains the evaluation of both normal and diseased case recognition comprehensively by reporting class-wise measures and the overall measures.

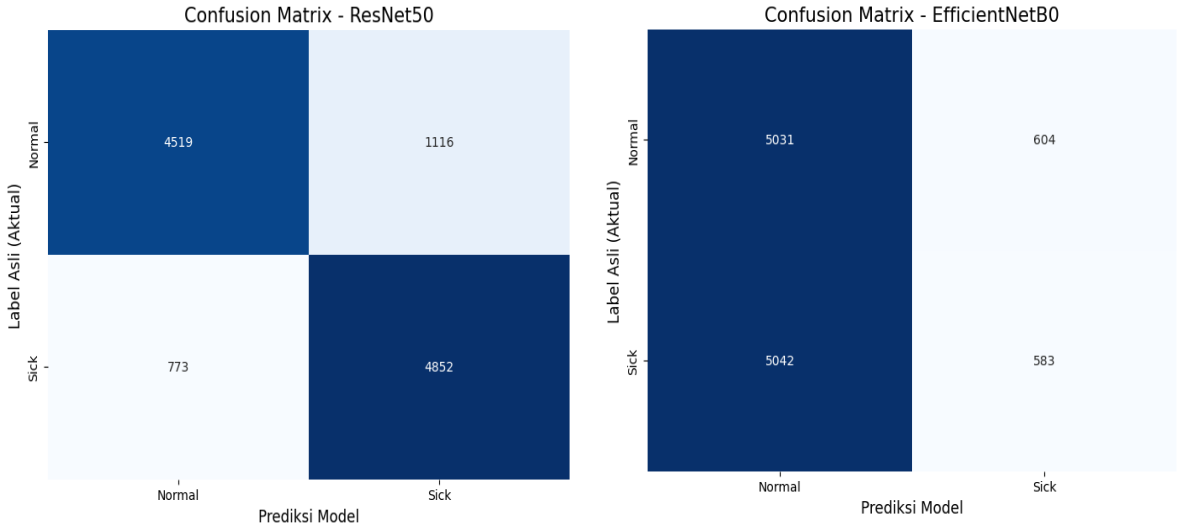
Table 4.3: Detailed Classification Performance of Individual CNN Models on the Test Set

Model	Class	Precision	Recall	F1-Score	Overall Accuracy	AUC-ROC
VGG16	Normal	0.93	0.95	0.94	0.9379	0.9877
	Sick	0.94	0.93	0.94		
ResNet50	Normal	0.85	0.80	0.83	0.8322	0.9191
	Sick	0.81	0.86	0.84		
EfficientNetB0	Normal	0.50	0.89	0.64	0.4986	0.5010
	Sick	0.49	0.10	0.17		
DenseNet121	Normal	0.95	0.95	0.95	0.9531	0.9930
	Sick	0.95	0.95	0.95		

The overall performance of DenseNet121 is the most favorable with balanced precision, recall, and F1-scores of 0.95 in both classes, which is why this model has a high overall accuracy of 95.31% and a great AUC-ROC of 0.9930. This means that it is highly discriminative and sensitive between classes. VGG16 is also doing very well with an

overall accuracy of 93.79 and a AUC-ROC of 0.9877 with well balanced class-wise metrics which indicates consistent generalization. ResNet50 achieves average results, with an accuracy of 83.22 per cent and an AUC-ROC of 0.9191, which indicates a fair level of classification but a relatively weak level of discriminative ability. As a contrast, EfficientNetB0 has much worse performance, an accuracy of 49.86, and an AUC-ROC value of the random guess, at least because the recall of the sick class is very low. By and large, the findings presented in Table 4.3 are quite sufficient to support the choice of DenseNet121 and VGG16 as the two most essential components of the ensemble model, as they underscore the relative inapplicability of EfficientNetB0 to the experimental environment.

Figure 4.3 shows the confusion matrices of the single CNNs, which are ResNet50, EfficientNetB0, VGG16, and DenseNet121, that were tested on the test data. Such matrices have a finer resolution of correct and incorrect predictions by a class, and provide more insight into the strength of each model to discriminate between normal and sick cardiac MRI images. The confusion matrices complement the quantitative measures that were reported above because they consider true positives, true negatives, false positives, and false negatives, thereby helping examine the clinical reliability of each model.



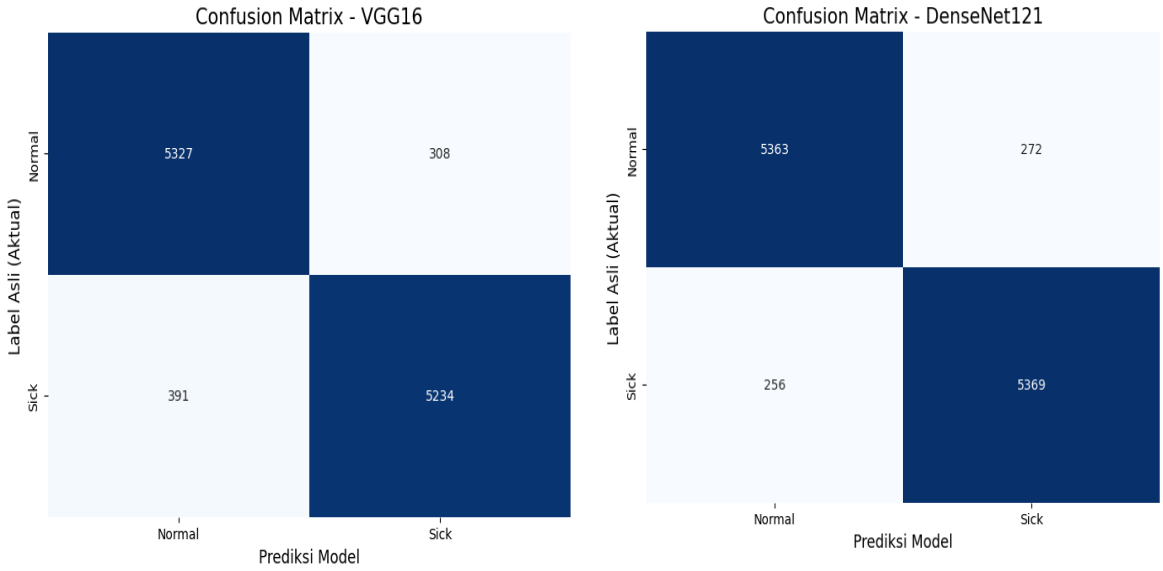


Figure 4.3: Confusion Matrices of Individual CNN Models

This is observed in the confusion matrix of ResNet50, which has quite a balanced classification performance and a large number of samples correctly classified as normal (4,519) and sick (4,852). Nevertheless, there is a visible amount of normal pictures being identified as sick (1,116) and some sick pictures are identified as normal (773), which is a sign of moderate confusion between the two classes. This performance is similar to mid-range performance of ResNet50 which has been shown to have good sensitivity but still has room to be improved with respect to minimizing misclassification errors.

On the contrary, the pattern of prediction of EfficientNetB0 is extremely skewed. Though it falsely categorizes many of the normal images (5,031), it falsely categorizes many of the sick images as normal (5,042), and its true positive classification of the sick class (583) is very low. This acute bias on the normal class corroborates the poor recall of diseased cases with the model and confirms that the model is not as ideal in reliable disease detection in the cardiovascular disease as in the specified training setup.

The VGG16 confusion matrix indicates that the classification behaviors are high, well-yielding, and balanced. The model correctly classifies most of normal (5,327) and sick (5,234) samples with rather low misclassifications. The few false negatives and false positives suggest that VGG16 has good capability to extract discriminative cardiac features, and it is a reliable model to be used in disease detection tasks.

DenseNet121 is the most successful model in terms of confusion matrix as it has the most correct predictions in the case of normal (5,363) and sick (5,369) categories and the lowest number of wrongly assigned classifications. This is due to the low number of false positives and false negatives that favor DenseNet121 over other models in terms of class

discrimination and balanced sensitivity that is important in a medical diagnosis scenario. On the whole, the confusion matrices in Figure 4.3 sufficiently support the choice of DenseNet121 and VGG16 as the most effective single models and are a strong argument to use them as the main participants of the proposed ensemble learning framework.

Figure 4.4 represents the curves of Receiver Operating Characteristic (ROC) of the separate CNN models: ResNet50, EfficientNetB0, VGG16, and DenseNet121, which are trained on the test data. The ROC curve shows the trade-off between the true positive rate and false positive rate at various classification thresholds whereas the Area Under the Curve (AUC) is one scalar measure of the overall discriminative performance. Each plot has the line on a diagonal which is a performance of a random classifier and is used as a reference point.

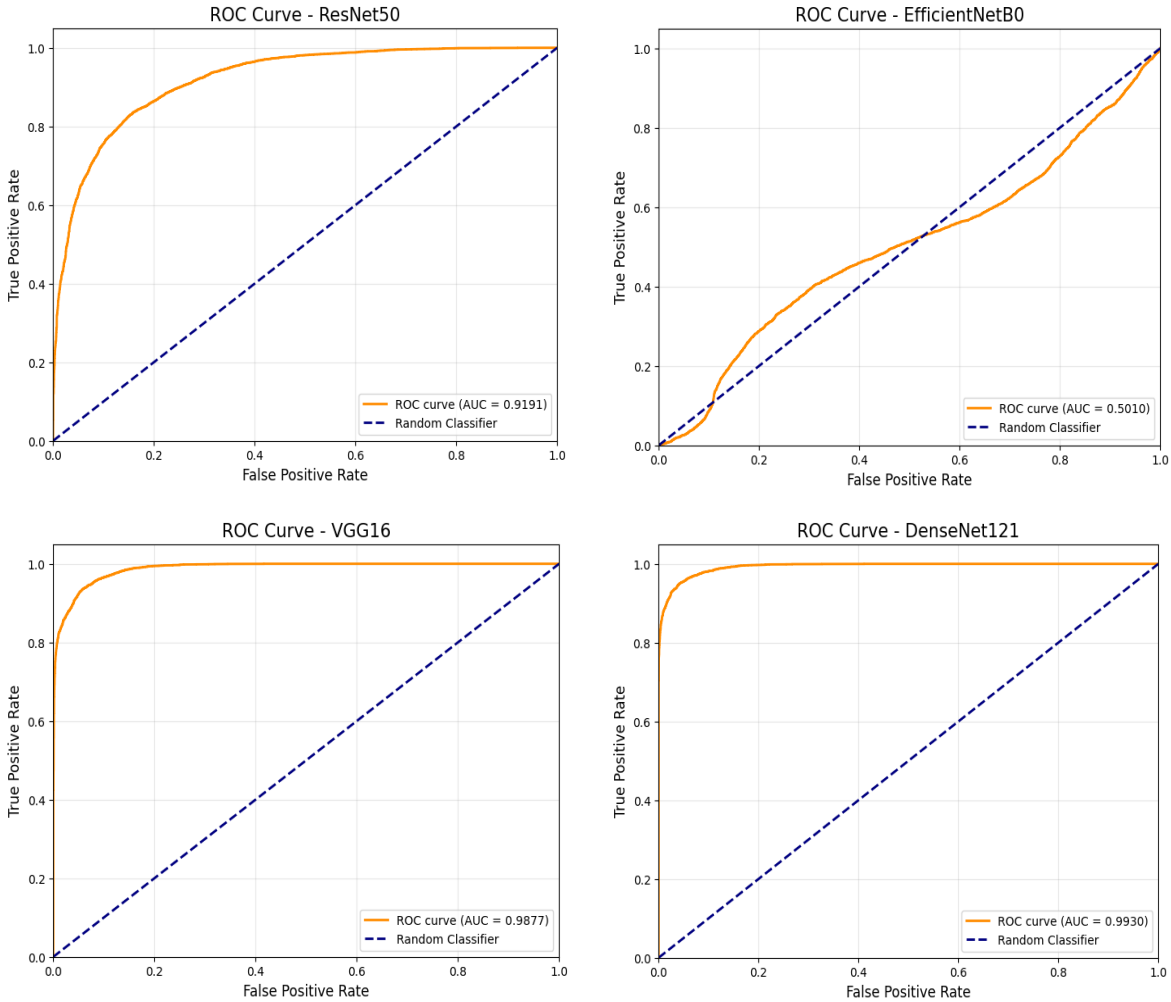


Figure 4.4: ROC Curves of Individual Models

The ROC curve of ResNet50 shows that it has good discriminative behavior with an AUC of 0.9191, which means that it is highly capable of differentiating normal and sick

cardiac MRI images. The curve is steeply increasing towards the upper-left end, which indicates a positive sensitivity-specificity ratio. On the contrary, EfficientNetB0 has a ROC curve that is very close to the diagonal base meaning an AUC of 0.5010. This outcome indicates the almost random performance of classification and proves the low efficiency of the model to differentiate the two classes when using the given training set up.

Both VGG16 and DenseNet121 demonstrate high ROC properties, and the curves of these two methods are directly near the upper-left part of the ROC space. VGG16 has a high classification reliability as it has AUC of 0.9877 and DenseNet121 has the highest AUC of 0.9930 which is a near-perfect classification. These two models have good generalization capability and robustness in cardiovascular disease detection as demonstrated by the consistently high ROC behavior of the two models. In general, the ROC analysis in Figure 4.4 once again contributes to the choice of both DenseNet121 and VGG16 as the most useful singular models and gives good reason as to why the two are used in the proposed ensemble learning design.

4.3.5 Ensemble Model Evaluation

A quantitative comparison of these ensemble learning strategies used in this study including average, weighted and voting based ensembles is detailed in table 4.4. These findings illustrate that when predictions of more than one high performing CNN model are combined, the overall classification performance is better than that of single models. Precision, recall, F1-score, accuracy, and AUC-ROC have been reported in the table to give a comprehensive analysis of the predictive strength and the ability of the classes to discriminate.

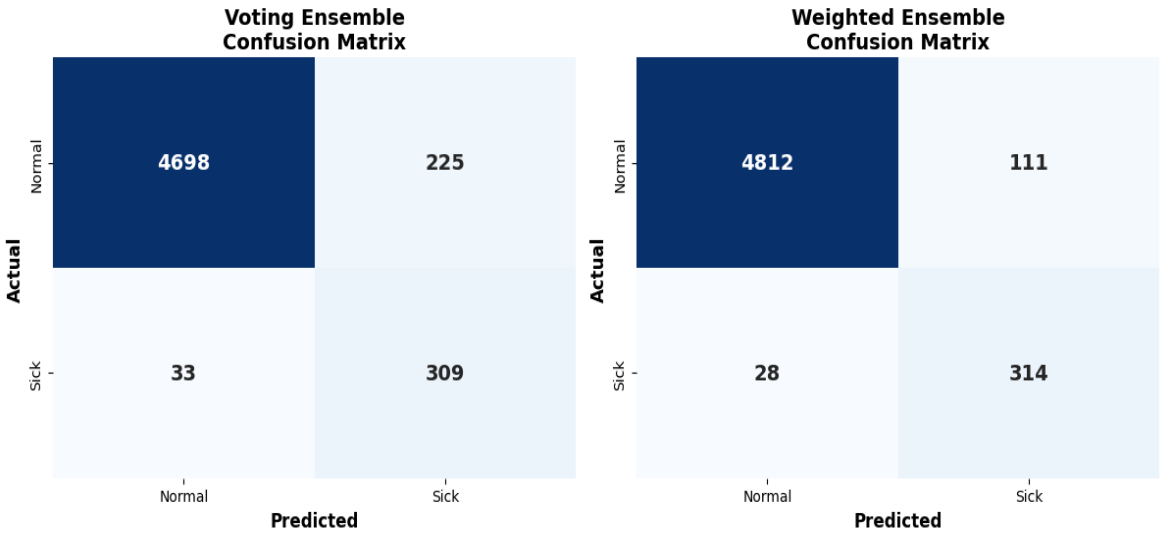
Table 4.4: Detailed Performance Metrics of Ensemble Models (Class-wise and Overall)

Model	Precision	Recall	F1-Score	Accuracy	AUC-ROC
Ensemble (Average)	0.7347	0.9152	0.8151	0.9730	0.9873
Ensemble (Weighted)	0.7388	0.9181	0.8188	0.9736	0.9876
Ensemble (Voting)	0.5787	0.9035	0.7055	0.9510	0.9289

The weighted ensemble is the best among the evaluated methods, having the highest overall performance with the accuracy of 97.36 and AUC-ROC of 0.9876. This is an indication of the success of applying the weight of the stronger base models, specifically

the DenseNet121 and VGG16, to ensure that the prediction of the base models can play a greater role in the final decision. The average ensemble is also a competent performer with an accuracy of 97.30% and a correspondingly high AUC-ROC, which demonstrates that even with a simple probability averaging, complementary information can still be obtained across multiple models. Conversely, the voting ensemble though with a high recall, recalls less precision and entire accuracy indicating a bias towards sensitivity at the expense of higher false positives. In general, Table 4.4 evidence conclusively that the weighted ensemble is the best and most balanced model, which is why it will be chosen as the final model to be deployed in the proposed cardiovascular disease detection system.

The confusion matrices of three ensemble learning methods, namely the voting, weighted, and average, are shown in figure 4.5 with respect to the test dataset being used. These matrices give a closer understanding of the classification effectiveness of each ensemble method in classifying normal and sick cardiac MRI images as they show the true positive, the true negative, the false positive, and the false negative. The results of the confusion matrix analysis are used to complement the quantitative measures of performance as it shows the actual effects of ensemble decision-making on classification errors.



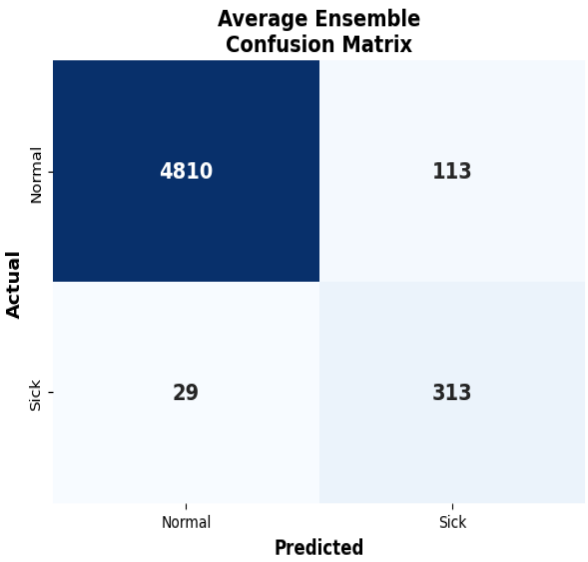


Figure 4.5: Confusion Matrices of Ensemble Models (Average, Weighted, and Voting)

The voting ensemble is very sensitive to the sick group and is able to detect a great number of sick cases well. Nonetheless, the technique has more false positives which are indicated by the number of normal images classified as sick. This response justifies why the voting ensemble has a high recall but relatively a low precision and overall accuracy. The average ensemble, on the other hand, has a more balanced pattern of classification, having fewer false positives and false negatives than voting. It is more sensitive to averaging probabilistic outputs of individual models which results to a better overall accuracy and greater consistency of the predictions.

The weighted ensemble has the best confusion matrix of all the ensemble methods. It rightly classifies the greatest number of both normal or sick samples and has the lowest misclassification rate. The fact that there are fewer false negatives is of great significance especially in medical diagnosis because it reduces the chances of false alarms. This ideal sensitivity-specificity ratio is the reason that the weighted ensemble has the highest accuracy and the largest value of AUC-ROC. In general, the confusion matrices of Figure 4.5 clearly indicate that weighted ensemble learning is more beneficial and has strong justification when it comes to the ultimate choice of the model to be implemented in the proposed cardiovascular disease detection system.

The comparison of ROC curves of the three ensemble learning strategies that include average, weighted, and voting strategies in the test dataset are shown in figure 4.6. The ROC curves show the trade-off between the true positive rate and false positive rate at variable classification thresholds and the straight line on the diagonal is the performance of a random classifier. Such comparison gives a clear visual evaluation of

the discriminative ability and strength of each ensemble method.

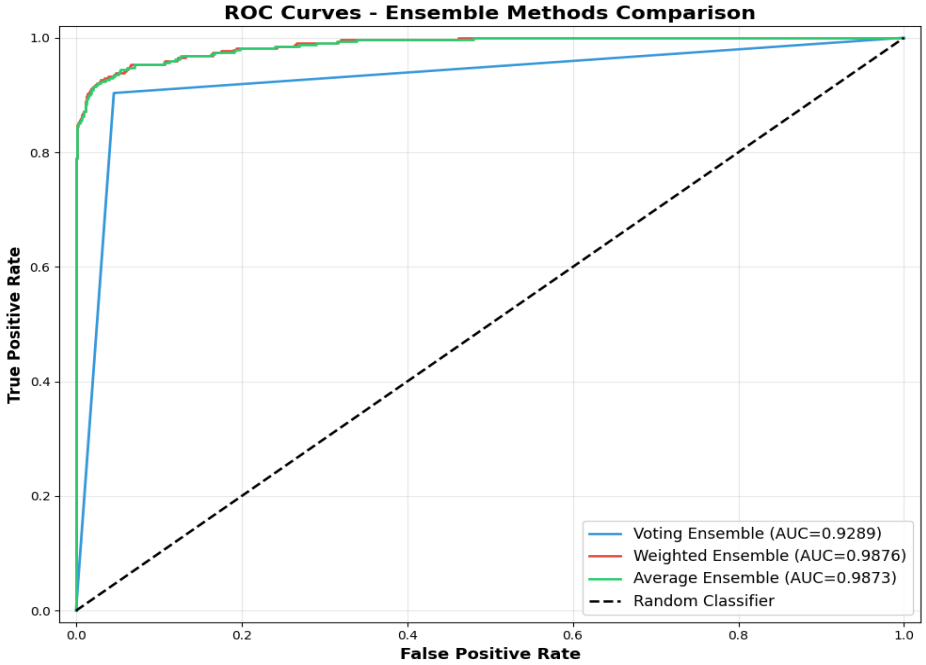


Figure 4.6: ROC Curves comparison of Ensemble Models (Average, Weighted, and Voting)

The weighted ensemble has shown the best ROC performance with an AUC value of 0.9876. It always has the closest curve to the upper-left corner of the ROC space, which means that it has an excellent sensitivity and specificity over a wide range of thresholds. The mean ensemble comes close and has an AUC of 0.9873, with similar discriminative ability and consistent classification. These findings confirm that the use of probability-based fusion strategies can quite successfully combine the advantages of single CNN models, especially DenseNet121 and VGG16.

Conversely, the voting ensemble has a relatively lower AUC of 0.9289 and ROC performance. It is still a much more flexible than a random classification, nevertheless, its curve demonstrates less flexibility at thresholds because it has a discrete decision mechanism. Comprehensively, Figure 4.7 shows clearly that the weighted ensemble offers the best and balanced discrimination ability, which supports its choice as the best ensemble model to be used in the detection of cardiovascular diseases and its application in the proposed system.

4.3.6 Comparative Analysis of All Models

Table 4.5 gives a full comparative view of the performance of all the experimented models, the individual CNN and the ensemble learning methods. The table provides a summary of the most important evaluation measures, such as accuracy, precision, recall,

F1-score, and AUC-ROC, which enables an effective evaluation of the enhancement of ensemble methods compared to the performance of individual models. The comparison, by comparing all the models in the same experimental conditions, shows the value of model diversity and fusion of predictions to the total diagnostic accuracy.

Table 4.5: Comparative Performance of All Models (Individual vs. Ensemble)

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
VGG16	0.9379	0.9444	0.9305	0.9374	0.9877
ResNet50	0.8322	0.8130	0.8626	0.8371	0.9191
EfficientNetB0	0.4986	0.4912	0.1036	0.1712	0.5010
DenseNet121	0.9531	0.9518	0.9545	0.9531	0.9930
Ensemble (Average)	0.9730	0.7347	0.9152	0.8151	0.9873
Ensemble (Weighted)	0.9736	0.7388	0.9181	0.8188	0.9876
Ensemble (Voting)	0.9510	0.5787	0.9035	0.7055	0.9289

DenseNet121 has the best performance that has 95.31% accuracy with AUC-ROC of 0.9930, which means high discriminative power and equal class coverage. VGG16 is also a fairly good model with an accuracy of 93.79 per cent and a high AUC-ROC of 0.9877 that indicates good generalization ability. ResNet50 is more resilient with bad performance, whereas EfficientNetB0 is poor with metrics that are nearly random guessing with the sick class almost at zero recall. These findings thus make it clear that DenseNet121 and VGG16 are the best individual contributors.

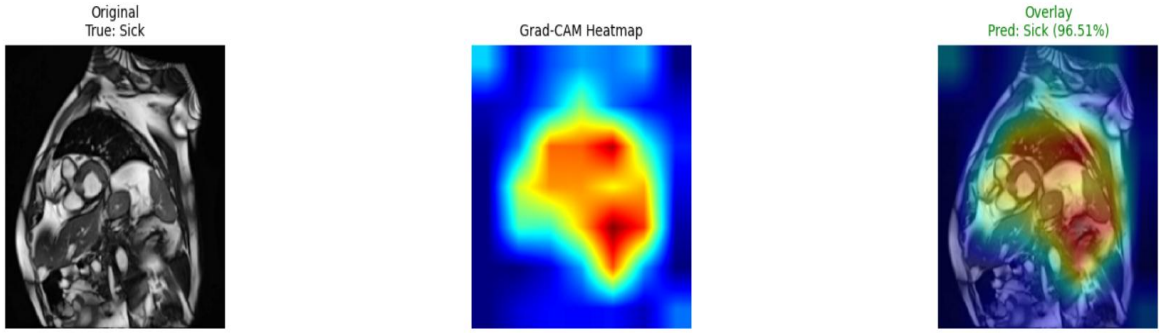
The ensemble models also improve the classification performance by combination of predictions of more than one strong CNNs. The average and the weighted ensemble method get the highest accuracies of more than 97% with the weighted ensemble slightly differentiating all other methods in most of the metrics. Although the voting ensemble has high recall, it has lower precision and total accuracy as a result of higher false positives. In sum, Table 4.5 shows that ensemble learning, especially, the weighted ensemble, can contribute greatly to robustness and predictive results, which confirms the choice of the final cardiovascular disease-detecting and deploying model.

4.3.7 Grad-CAM Visualization Analysis of Individual CNN Models

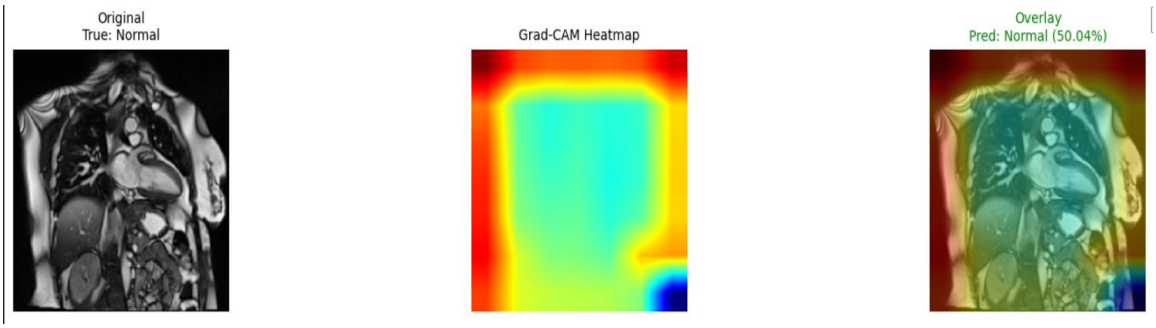
Figure 4.7 gives a comparative visualization of the results of Grad-CAM on the single CNN models ResNet50, EfficientNetB0, VGG16, and DenseNet121 to determine how each model perceives cardiac MRI images when making a prediction. The sets of images contain the original image of MRI, the necessary Grad-Cam heatmap, and the overlay

visualization of the regions that have most heavily contributed to the choice made by the model. These visual accounts give a critical understanding of the mechanisms of spatial attention of the models and it contributes to the interpretability of the proposed system.

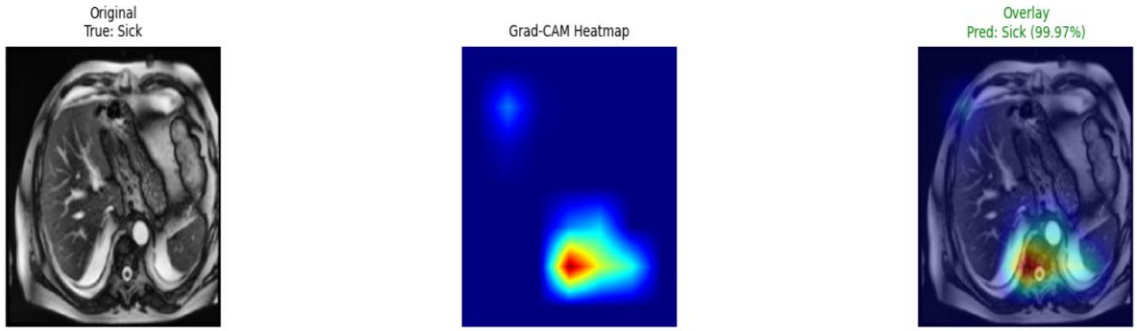
ResNet50



EfficientNetB0



VGG16



DenseNet121

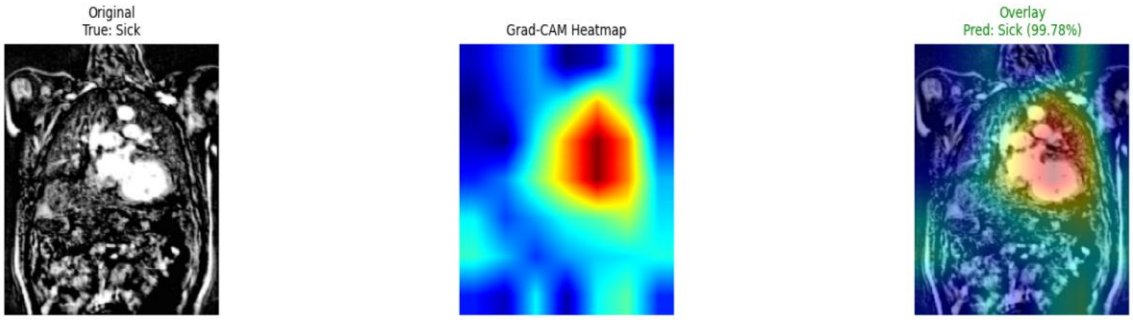


Figure 4.7: Comparison of Grad-CAM Visualization for Individual Models.

The ResNet50 Grad-CAM visualization reveals moderate localization of discriminative regions in the cardiac area in cases of sickness and the activation is mostly localized to areas of the anatomy. Although the model with the required confidence identifies the areas of disease relevance, the attention areas are relatively smeared, which means that ResNet50 does not rely on more focused patterns but the contextual ones. This finding is in line with its mediocre classification performance and occasional misclassifications in previous findings.

Conversely, EfficientNetB0 has fewer focused and inconsistent patterns of activities. The Grad-CAM heatmaps reveal that the attention is scattered or weak through the image even when the sample is correctly classified, which indicates that the model is not very good at finding the meaningful disease-related features consistently. This without proper localization justifies the poor recall and almost random performance of EfficientNetB0, which does not focus on clinically significant cardiac locations when making a prediction.

VGG16 Grad-CAM results show good and localized activation in areas of cardiac structures, especially with sick cases that have a high prediction confidence. The heatmaps suggest that VGG16 can target disease-relevant areas of the body, and is therefore a reliable and interpretable predictor. This obvious spatial focus underlines the high quantitative abilities of VGG16 and the ability of the model to identify discriminative visual patterns in cardiac MRI scans.

DenseNet121 is the only model that has the most accurate and focused Grad-CAM visualizations. The activation maps are highly localized to the regions that are of the key to the presence of the disease and little is paid to the irrelevant background regions. This localization is high-quality, which results from the better reuse of features and deep connectivity of DenseNet121 and allows it to learn highly discriminative and clinically meaningful representations. Altogether, Figure 4.8 Grad-CAM comparisons justify the quantitative results and prove that DenseNet121 and VGG16 not only have higher accuracy, but also provide better interpretability and could be used effectively as a part of the final ensemble-based cardiovascular disease detection system.

4.4 Summary

This chapter involved an experimental assessment of the offered cardiovascular disease detecting scheme using the methods of deep learning and ensemble learning. The results of individual CNN models (ResNet50, VGG16, DenseNet121, and EfficientNetB0) were systematically examined in the form of various metrics of evaluation, such as accuracy,

precision, recall, F1-score, ROC, and confusion matrices, and training behavior. The findings have shown that there are clear performance disparities in the models with the best discriminative ability being shown by DenseNet121 and VGG16 and the limitations of EfficientNetB0 are revealed in the specified experimental conditions. Moreover, the approaches of ensemble learning were tested to unite the merits of the most successful separate models. The weighted ensemble was the most effective of the tested methods and it provides the best trade-off between the sensitivity and specificity. Grad-CAM visualizations were found to be significant to model decision-making as they were useful in identifying clinically relevant cardiac areas, thus improving interpretability and reliability. In general, this chapter confirmed the usefulness of the suggested ensemble-based method and formed a solid empirical basis of the ultimate deployment of the system in a mobile application on detecting practical cardiovascular diseases.

Chapter 5

Engineering Standards and Design Challenges

The chapter talks about the engineering principles, standards, and practical constraints applied in the entire design of the cardiovascular disease detection system. It points out the software, hardware and communications standards which informed the design and implementation processes. Another theme outlined in the chapter is the societal, ethical, and sustainability impacts of the project, the roles and duties to be assumed in the development of medical AI systems. It looks at the complicated engineering issues facing it and how they were solved by means of a coherent thinking and technical judgments. Moreover, it generates the mapping of the project to known activities in engineering to present that the project is being done in accordance with the professional and academic standards. Collectively, these discussions indicate that not only is the system technically sound but it is also ethically responsible and in tandem with modern engineering practices.

5.1 Compliance with the Standards

The design of this cardiovascular disease detection system is in accordance with the necessary engineering and software requirements that will guarantee reliability, safety, and reproducibility. During the project, the common data processing, training models, and software development principles were used, such as regular data system organization, keeping track of files, and following ethical standards regarding handling of medical photographs. Its implementation is done by following structured codes, versioning, and documented workflow in order to ensure transparency and traceability. Fair and meaningful model assessment was achieved through the use of standard measures of evaluation like accuracy, precision, recall, F1-score, and ROC-AUC. Further, the system architecture conforms to the typical guidelines of deep learning by relying on established libraries like TensorFlow, Keras, and scikit-learn, and by employing the well-known techniques like transfer learning, regularization, and early stopping. Ensemble learning also helps to achieve a high performance level which decreases the bias in the model and improves the generalization as per the best practice

in the AI model design. The project as a whole is in line with the engineering and software standards to provide a reliable and ethically responsible diagnostic solution.

5.1.1 Software Standards

The software development phase of this project will be done under the guidelines to achieve uniformity, dependability, and sustainability of the entire system. The coding practices are well structured such the implementation adheres to structured coding practices such as modular design, documentation and using version control to trace updates and maintain the integrity of the code. Recognized industry libraries like TensorFlow, Keras, scikit-learn, NumPy and Matplotlib were utilized so that they are compatible, long term support and following well accepted machine learning frameworks. Procedures related to data preprocessing, training a model and evaluation were run through pipelines that were reproducible, featuring fixed list of random seeds and standard dataset manipulation procedures. Moreover, the project is developed with best practices and the application of deep learning development concepts, such as the use of callbacks, checkpoints, and optimization techniques, to guarantee model efficiency and stability. These software standards allow the study to make sure that the system can be replicated, extended and evaluated to depend on by other researchers or practitioners.

5.1.2 Hardware Standards

The hardware guidelines used in this project guarantee that the system is created and put to test in a dependable, efficient and scalable computing setting. The experiments were conducted with hardware setup, which is usually used in deep learning research, such as GPUs to speed up training, enough RAM to work with large batches of MRI images, and stable storage to cope with datasets and model checkpoints. These hardware selections are in line with the best practice of training CNN and ensemble models that need high computational power to compute thousands of images, do augmentation, and get deep feature representations. The training process is accelerated by using the platforms that are equipped with a GPU and would also make sure that the models would be trained to work in realistic conditions that are similar to clinical AI systems. The project will ensure that other hardware standards are met, these standards ensure consistent performance and can be used to support reproducibility and to allow the possible deployment onto more advanced or cloud-based infrastructures in the future.

5.1.3 Communication Standards

The communication norms that are adhered to in this project are clear, consistent, and easily available to both technical and non-technical stakeholders in the flow of information, documentation, and reporting of the outcomes. All the methodological steps such as the preparation of the data, the training of the model, the evaluation and the integration of the ensemble were recorded through structured comments, descriptive names of variables and through organized sections in the notebook. Well-labeled graphs, confusion matrices, ROC curves, and sample prediction outputs were also upheld to support the practice of visual communication and therefore, make it easy to interpret the behavior of the system. Results were presented in standardized reporting formats so that the values of accuracy, precision, recall, F1-score, and AUC are presented in a clear and meaningful way. Besides, the thesis is written according to the rules of the academic writing in order to present ideas about deep learning and medical imaging in a direct way. These standards of communication aid in establishing transparency, reproducibility and clarity which facilitates the design of the system and its results to be perceived and judged suitably.

5.2 Impact on Society, Environment and Sustainability

The implications of the development of an automated cardiovascular disease detection system are significant in the sense of the society, environment, and sustainability in the long term. The system can assist in mitigating the workload of medical professionals, increase the rate of early diagnoses, and eventually lead to improved patient outcomes, particularly the ones who have little access to medical specialists in the community. MRI-based deep learning tools are also useful in facilitating more effective utilization of the hospital resources by eliminating redundant tests, and patient waiting time. Environmentally, digital diagnostics allow decreasing the use of physical resources and the necessity to perform several imaging processes, thereby decreasing energy consumption and resource use related to medical equipment. Regarding sustainability, the project fosters scalable and reusable AI practices through transfer learning and ensemble schemes, which, in other words, means that the same framework can be used across other medical imaging tasks without making major resource requirements. In general, the proposed study will lead to a fairer, more efficient, and more ecologically friendly healthcare system through the implementation of the most recent deep learning technologies in the diagnostic practice.

5.2.1 Impact on Life

The proposed machine learning-based cardiovascular disease detection system can enhance the life of a human being to a great extent by making the diagnosis of heart-related conditions possible earlier and more accurate. Early diagnosis would lead to serious complications and decrease hospitalization rates and improve the long-term health outcomes of patients. The system can help medical professionals to detect abnormalities more effectively since it automates the process of analyzing MRI images, which is particularly useful in the environment where the level of specialists knowledge is lower, or the amount of patients is enormous. This will minimise chances of missed diagnoses and aid in more uniform clinical decision making. The technology also has the power of empowering people in distant or underserved areas because advanced diagnostic support is made more accessible under digital healthcare platforms. The net effect of the system is enhanced health management, quality life, and enhanced survival of individuals who are at risk of cardiovascular disease.

5.2.2 Impact on Society & Environment

The application of the deep learning to cardiovascular disease prediction has a positive impact on society as it facilitates the accessibility of healthcare, aids clinical decision-making, and minimizes the difference in the quality of diagnostic services in the various areas. Automated MRI analysis assists hospitals in handling the growing patient number in a more effective way, relieving medical personnel and allowing making treatment decisions faster. This might be particularly useful in cases with low resources when there might be no specialist cardiologists or radiologists easily accessible to those communities. Digitally, digital diagnostic tools help cut the energy usage and material waste on the facilities of health care institutions through dependence on repeated imaging and physical documentation. The system also reduces unwarranted follow-up scans by facilitating more precise first-time diagnosis and, this way, results in more sustainable medical equipment and resource use. In general, the project aids the more efficient, fair and eco-friendly healthcare ecosystem.

5.2.3 Ethical Aspects

Ethics is a factor that will be significant in designing an automated cardiovascular disease detector system, particularly in a medical image and patient-related image. In this project, there is a focus on responsible data processing with publicly available and anonymized MRI data being used so that no personal or sensitive data are revealed. The

system will support, not substitute, the clinicians and will help to avoid excessive dependence on automated decisions and preserve the central position of human judgment in the medical profession. They have ensured that they have avoided bias by ensuring that even-handed data preprocessing and strict evaluation procedures have been administered, to minimize chances of unfair or inaccurate predictions of specific groups of patients. Visualization tools are also encouraged like t-SNE plots and the prediction outputs to enable users gain a more insight into the way the model comes to its conclusions. The ethical standards that are upheld by this project by prioritizing privacy, fairness, accountability, and transparency are the keys to fostering trust in AI-assisted healthcare systems.

5.2.4 Sustainability Plan

The sustainability plan of this cardiovascular disease detection system is based on the long-term usability, flexibility, and the efficiency of the resources usage. With transfer learning and a modular structure of deep learning, the system can be adapted or enlarged with only slight extra training expenses, which is why it can be applied to future datasets or novel heart diseases. The ensemble structure is more stable, and does not require frequent retraining, which leads to a decrease in the number of computations required in the long run. Code and workflows are both open-source and reusable and scalable, like hospital systems or cloud-based medical systems. Also, the use of digital diagnostics is in line with sustainable healthcare as it minimizes paper-based reporting, minimizes the repetition of imaging, and encourages energy-efficient clinical processes. This will help in the longevity of the technological use as well as responsible usability of resources so that the system in the long run is practical and environmental responsible.

5.3 Project Management and Financial Analysis

To successfully perform this research on cardiovascular disease detection based on deep learning and ensemble-based CNN models, the key aspects that were necessary were effective project management and financial planning. Since the analysis of medical images requires accuracy, computational effectiveness, and rigor, the project was well designed to guarantee systematic improvement, repeatability, and economic sustainability. This section will describe the structure of the workflow that was followed between the collection of the data and the deployment of models and how the financial aspect was handled to make it still a viable project without affecting the quality of the research.

5.3.1 Project Planning and Task Management

The system development was guided by a more structured, phase-based management approach, which was inspired by agile practices. Every stage came along with specific output, milestones, and chances of improvement. The key steps were acquisition of datasets, preprocessing and augmentation, development of CNN model, ensemble integration, evaluation, interpretability and final reporting.

This project started with finding and realising of the MRI dataset, checking the quality of images, and sorting between the Normal and Sick classes. The preprocessing steps involved image resizing, pixel value normalization, augmentation and the 70:20:10 train-validation-test split to balance the performance evaluation. Having the dataset prepared, four ready-to-use CNN models, namely VGG16, ResNet50, EfficientNet, and DenseNet, were configured with their own customized classification layers and each trained separately on early stopping and learning-rate schedule.

The second step was to test these models in terms of accuracy, precision, recall, F1-score, AUC, confusion matrices, and ROC analysis. Ensemble methods including simple average, weighted average, and majority voting were subsequently applied to enhance the predictive stability and minimize individual-model biases. t-SNE visualization and sample prediction results were subsequently adopted to understand how the model behaves and prove that the features are separable. Every process was recorded in order to promote transparency and reproducibility that were backed by version control and tracking of the experiment through GitHub, Google Drive, and Jupyter Notebooks.

1. Resource and Time Allocation

The project was done within a time span of about 5 months and it was organized as follows:

- Month 1: Data gathering, visual image analysis, plan of image augmentation, and research literature.
- Month 2: Preprocessing, augmentation, splitting of data, and setting up of environment.
- Month 3: Model training and development (VGG16, ResNet50, EfficientNet and DenseNet)
- Month 4: Virtue design, evaluation of performance, visualization of Grad-CAM.
- Month 5: Visualization of the prediction, documentation and writing of thesis.

The application of Google Colab with a GPU acceleration reduced the costs of computing and allowed training large models without having specific hardware.

2. Financial Analysis

One of the strengths of this study is its cost-effectiveness. The majority of the components, such as datasets, software libraries, and training platforms, were open-source and free. Google Colab offered free-of-charge training platforms that had GPU support, thereby saving a lot of money in terms of computational costs. Due to such financial benefits, the project is viable to academic research and can be expanded to larger clinical trials. The deployed final system is accessible in the resource-constrained areas because it can be installed in common workstations, or in cloud environments without requiring licensing costs.

3. Scalability and Future Investment

The modularity and scalability of the methodology were taken into account. All its components, such as preprocessing, single CNN models, ensemble strategies and evaluation tools, can be extended to larger datasets or other types of diseases. Possible future investments might be involved in paid cloud computing that could be used to train faster, integration with clinical PACS systems, or a lightweight mobile-based diagnostic tool that would be used to perform real-time MRI analysis. In the present design, a cost-effective, replicable, and deployable framework is already shown, which can be applied to academic research and even in the real healthcare environment. The project was developed in a multi-stage format, that is, it entailed:

- System design and requirement analysis.
- Preparation and augmentation of data sets.
- Development and training CNN model.
- Teams merging and performing self-assessment.
- The visualization of interpretability and prediction.
- Record keeping and reporting.

5.3.2 Tools and Platforms Used

Table 5.1 provides a summary of the key tools and platforms applied during the project.

Table 5.1: Details of Tools and Platforms Used

Category	Tool/Platform	Purpose
Programming Language	Python	Core development and preprocessing
Libraries/Frameworks	TensorFlow/Keras, scikit-learn, NumPy, Matplotlib	Model training, evaluation, visualization
Model Training	Google Colab (GPU)	Training CNN and ensemble models efficiently

IDE	Jupyter Notebook	Experimentation, analysis, and workflow tracking
Version Control	Git/GitHub	Code versioning and collaboration
Storage/Backup	Google Drive	Dataset storage and experiment backup

5.3.3 Financial Analysis

This study was crafted to be easy to access and economically viable to students and single research studies. The approximate cost breakdown is illustrated below.

Table 5.2: Estimated Cost and Financial Analysis

Resource/Item	**Estimated Cost (BDT)**	**Remarks**
Google Colab Pro (optional)	3500	Useful for longer GPU sessions
Computer (personal/lab)	Existing Resource	Used for implementation and testing
Internet Access	4200	Required for cloud-based training
Python Libraries/Frameworks	0	All open-source
Cloud Storage Backup	300	Optional (Google Drive)
Total	8000 BDT	Approximate cost

5.4 Complex Engineering Problem

An automated cardiovascular disease detection system built on deep learning has various levels of complexity. All these complexities are due to the processing of medical imaging data, the design and training of state-of-the-art CNN and ensemble models, considering ethical and engineering requirements, and the attainment of credible diagnostic accuracy. In this section, the project is analyzed in light of the definition of a complex engineering problem, mapping the project to the existing knowledge categories, such as problem-solving categories and engineering activities.

5.4.1 Complex Problem Solving

This subsection gives the mapping of the research problem with the seven criteria of the Complex Engineering Problem (EP). The mapping is formed in Table 5.3 then there are concise reasons in bullets.

Table 5.3: Mapping with Complex Engineering Problem.

EP1 Dept of Knowledge	EP2 Range Of Conflicting Requirements	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent Of Stake-holder Involvement	EP7 Interdependence
✓	✓	✓	✓			✓

Justifications:

- EP1: Depth of Knowledge - The solution must have deep learning, transfer learning, ensemble learning and medical image analysis knowledge. As Chapter 3 (Methodology) outlines, various CNN structures (ResNet50, VGG16, DenseNet121, EfficientNetB0) are trained and adapted and Chapter 4 (Results and Discussion) evidences the details of their performance analysis through various measures. This is more knowledge than the simple fundamentals of engineering.
- EP2: Range of Conflicting Requirements - The proposed system strikes a balance between the competing needs, including high diagnostic accuracy, computational efficiency, model interpretability, and the capability to run the models on a mobile platform. Chapter 3 is a trade-off involving the choice of model design and ensemble strategies, and Chapter 4 is a performance improvement evaluation without undue sophistication.
- EP3: Depth of Analysis - The problem requires much analytical experimentation, such as the training of behavior, confusion, ROC curves, and comparison in ensembles. Chapter 4 gives a detailed discussion of the individual and group models, but there are profound analytical insights that are made, not just a facade assessment.
- EP4: Familiarity of Issues - Cardiovascular disease detection using cardiac MRI images is not a standard engineering problem. In Chapter 3, the issues of image variability, disease representation, and class imbalance are mentioned, and it is necessary to adjust existing CNN models to a specific field of medicine.
- EP7: Interdependence - There is strong interdependence between the system components. The preparation of datasets, training of the model, its integration into an ensemble, and interpretability analysis, as well as the deployment to mobile devices are closely related as depicted by the methodological flowchart in Chapter 3 and confirmed by experimental evidence in Chapter 4.

Mapping with Knowledge Profile

Table 5.4: Mapping with knowledge Profile.

K1 Natural Science	K2 Mathematics	K3 Engineering Fundamentals	K4 Specialist Knowledge	K5 Engineering Design	K6 Engineering Practice	K7 Comprehension	K8 Research Literature
		✓	✓	✓	✓		✓

Justifications:

- K3: Engineering Fundamentals - The system is based on the standard principles of machine learning and image processing, which underlie both model training and evaluation, as is described in Chapter 3.
- K4: Specialist Knowledge -DenseNet121 and VGG16 were chapters 3 and 4 respectively. Advanced Knowledge of deep CNN architectures and transfer learning is necessary to implement and compare such models.
- K5: Engineering Design - Chapter 3 contains the entire system design, which comprises of preprocessing, model selection, ensemble strategy, and deployment planning.
- K6: Engineering Practice - Chapter 3 and Chapter 4 cover issues of practical implementation, including configuration of training, and hardware dexterity and adjustment, as well as execution optimization.
- K8: Research Literature - The choice of models, methodological approach, and evaluation standards follows the precedent research, and it is discussed in Chapter 2.

5.4.2 Engineering Activities

The section intersects the research work to the categories of Complex Engineering Activities (CEA). It is presented in the map in Table 5.5 and is followed by brief reasons.

Mapping with Complex Engineering Activities

Table 5.5: Mapping with Complex Engineering Activities.

EA1 Range of resources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓		✓	✓	

Justifications:

- EA1: Range of Resources - The project involves the use of big MRI data, deep learning systems, and mobile deployment systems, as outlined in Chapter 3.
- EA3: Innovation - The ensemble learning with Grad-CAM interpretability in cardiac MRI classification is an innovative technique as it is shown by the comparative results in Chapter 4.
- EA4: Consequences for Society and Environment - The proposed system can improve patient outcomes and reduce healthcare burden since early detection of the disease can be performed. This social influence is reflected in Chapter 1 and 6.

5.5 Summary

In the chapter, the main engineering criteria and design issues that were taken into account in designing the cardiovascular disease detecting system were described. It also talked on the adherence to software, hardware, and communication standards so as to maintain reliability, reproducibility, and clarity throughout the project. It was also seen in the chapter how the system in general affected human life, society, environment and long-term sustainability with how automated MRI analysis would help in early diagnosis and responsible use of healthcare resources. Such ethical factors like privacy of data, justice, and transparency were emphasized to make sure that this system fits into the responsible AI practices. Combining these considerations, it can be seen that the project was created with a keen interest not only on technical performance, but on the societal value, ethical responsibility and sustainable design.

Chapter 6

Conclusion

This chapter is an overview of the major findings of the study, and a reflection of the general effectiveness of the suggested deep learning system. It brings to the fore key contributions of the work such as the effectiveness of ensemble learning in enhancing MRI based cardiovascular disease detection. The chapter also touches on the limitations that were experienced in the study and how the limitations affected the findings. Suggestions on the future research directions have been made as an extension of the capabilities of the system to make it a more clinically relevant system. The chapter provides these insights to draw the thesis into a logical conclusion and underline the importance of the work in the development of medical AI applications. It confirms the possibilities of the system as the basis of additional innovation and real-life implementation.

6.1 Summary

The thesis has introduced a deep learning-based cardiovascular disease detection model using cardiac MRI images, specifically the design and testing of four transfer learning CNN models: ResNet50, VGG16, DenseNet121, and EfficientNetB0, and three ensemble learning methods. The authors utilized a systematic approach of the methodology pipeline that involved dataset preparation, preprocessing, data augmentation, and systematic training of the individual CNN models in a unified experimental condition. The results of the most efficient models were combined by ensemble methods, i.e. average, weighted and voting, and assessed on the basis of various performance measures. As shown in the conducted experiments, ensemble learning and especially the weighted ensemble method performed more stable and accurate predictions when compared to the individual CNN models. The interpretability of the models was further improved by using Grad-CAM, which gave a significant insight into the decisions that the model made on the unknown test images. Lastly, deployment of the optimized ensemble model was also presented in a mobile application, which shows the practicality of the proposed system in the context of real-time and accessible cardiovascular disease detection. In general, the research presented supports the idea that the hybrid use of a set of pretrained CNN structures can substantially enhance the diagnostic reliability

and contribute to the early diagnosis of the disease in clinics and in studies.

6.2 Limitation

Though the proposed cardiovascular disease detection system has risen well, there are various limitations that must be considered in accordance with the methodology adopted. A major limitation is connected with the size and variety of the data set. This cardiac MRI dataset utilized in this paper is publicly accessible and well-labeled, but it may not be as representative of the overall versatility of clinical real-world scenarios, including variation in patient demographics, MRI equipment, acquisition protocols, and imaging condition. This could then influence the generalization ability of the trained models when they are implemented on bigger and more diverse clinical data. Although data augmentation has been used to minimize overfitting and make the models more robust, they cannot completely eliminate the necessity of different samples in the real world.

The other weakness is related to the cost of computation of the proposed framework. Combining various transfer learning-trained CNNs, including ResNet50, VGG16, DenseNet121, and EfficientNetB0, with various ensemble strategies also leads to an increase in the amount of training and inference. Despite the fact that training time under transfer learning is lower than when training is done under the scratch training method, the total computational requirement is still high, especially during model training and ensemble integration. This can restrict the ability to scale the system to institutions or a medical environment with relatively small amounts of GPU resources and can be problematic in time-sensitive clinical deployment use cases.

The aspect of interpretability is also one that can be improved. Although Grad-CAM was used to explain model predictions visually and offer greater transparency, the given interpretability analysis was restricted to a set of models and examples. The clinical trust and decision support could be further reinforced through more comprehensive and systematic explainability techniques. Lastly, the system is set to work with binary image-level classification (Normal vs. Sick), and this feature limits its capacity to detect particular cardiovascular diseases or evaluate the severity of the disease. The framework could be expanded to include multi-class classification to enhance its clinical applicability with the help of more detailed diagnostic information.

6.3 Future Work

Based on the proposed cardiovascular disease detection framework, a number of meaningful extensions can be made based on the approach taken. The extension of the

dataset with more and more diverse set of cardiac MRI images obtained in various hospitals, in various MRI scanners, and in different patient groups can be considered one of the significant directions of the future work. This diversity would result in a better generalization ability of the models and increase their applicability in clinical settings in the actual world. Also, a multi-class classification system (Normal vs. Sick) should be extended to the existing binary type of classification system, which would enable the system to estimate the specific cardiovascular or disease stage, which would be more useful in clinical practice and inform the well-informed medical decision-making.

The next research area that may be considered a promising move is searching deeper deep learning structures and en semblance strategies. Whereas this experiment was on the CNN models based on transfer learning like ResNet50, VGG16, DenseNet121, and EfficientNetB0, future experiments may include transformer-based models, CNN-Transformer architectures, or attention mechanisms to enhance feature representation in cardiac MRI images. Further, other ensemble methods, including stacking, meta-learning, or knowledge distillation, might be explored to improve the performance at less complex computational costs. Regarding interpretability, future research can combine more explainable AI techniques with Grad-CAM to obtain a more in-depth and clinically significant elucidation of model predictions.

Lastly, the expansion of the mobile application and deployment framework is one more very important step to the real implementation. The directions that the future may pursue are the optimization of models performance to work with real-time inference, the design of the user interface, and changing the system to cooperate with the hospital information systems or clouds. The system should be validated by consulting medical professionals and testing it in real clinical processes, as this would give useful feedback to optimize the system in accordance with the practical diagnostic needs and make the use of the system in healthcare facilities safe and reliable.

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